

Data Visualisation

Week 1 - Introduction

Dr. Bojan Božić

Credits

- This course is based on material developed by :
 - Cathy Ennis
 - Marisa Llorens-Salvador
 - John McAuley
 - Brian Mac Namee
 - Colman McMahon
- Material from other sources is also used (citations included)

Overview

- Preliminaries
- Visualisation
- Module Overview
- Books
- Tools
- Video -The Joy of Stats

Learning Outcomes

- Demonstrate understanding of how humans perceive the world around them on a general level and absorb complex data/information on a specific level.
- Analyse and evaluate how metaphors are used to convey unfamiliar information.
- Analyse and evaluate how mental models aid in the interpretation of complex visual displays.
- Select, formulate and integrate metaphors to suit data-driven tasks
- Demonstrate an understanding how visualisation is used in data journalism to communicate complex ideas and stories
- Demonstrate understanding how visualisation is used in story telling

PRELIMINARIES

My Details

- Name: Bojan Božić
- Email: bojan.bozic@tudublin.ie
- Brightspace (Notes, Labs, Assignments):
 - Module Name: Data Visualisation SPEC9995: 2025-26

Course Delivery

- Full Time TU059
 - Lectures/Labs: Tuesday 9:00 – 12:00
 - On Campus Lectures and Labs
 - Every week we will first have the lecture (the duration will vary depending on the topic) and the lab right afterwards.
 - You will also be able to present your work during the lab.

Assessment

- Weekly Labs (10%):
 - 8 labs in total
 - 5 of these labs are worth 2% each
- Assignment 1 (30%): Visualisation project using Tableau
- Assignment 2 (30%): Project using R
- Exam (30%)

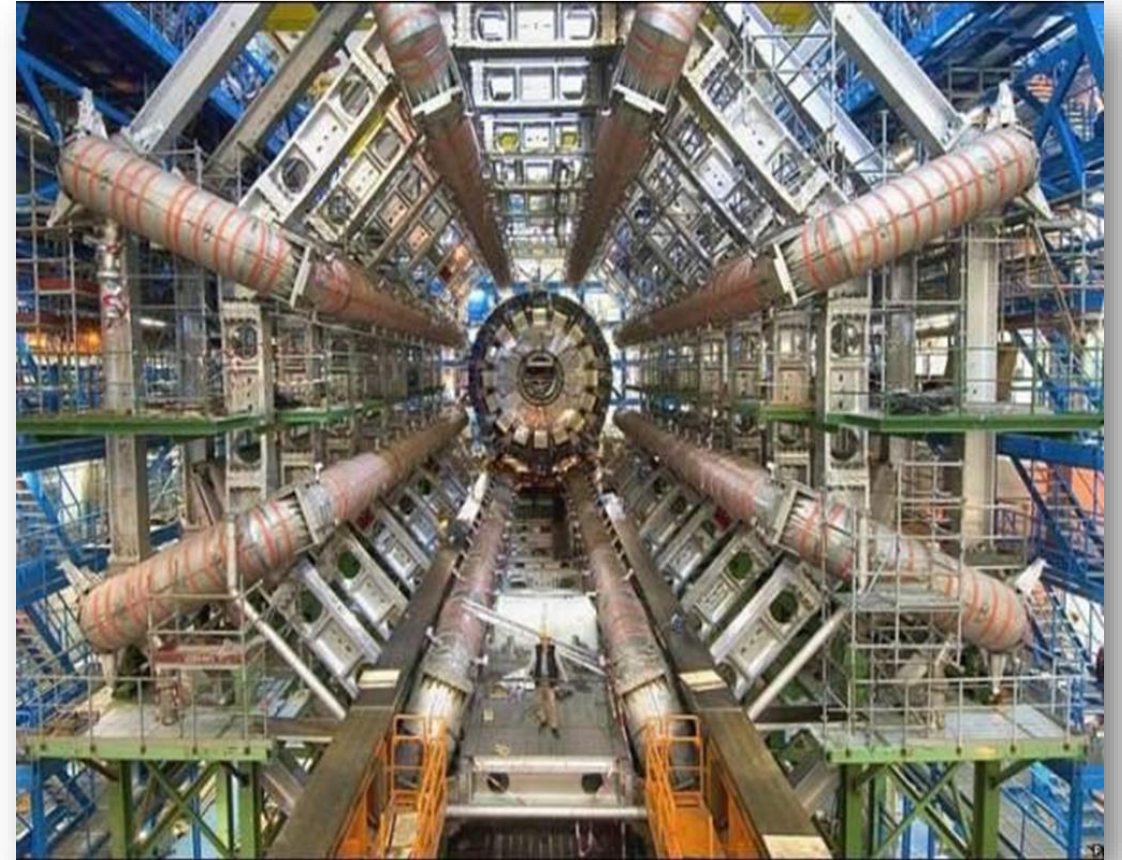
Course Overview

Date	Week	Lecture	Learning Outcomes	Lab Number	Lab Topic	Lab Marks	Assignment Specified	Assignment Due
16/09/25	1	Introduction	LO1, LO2, LO3, LO4, LO7, LO8					
23/09/25	2	Categorisation	LO1, LO2, LO3, LO4, LO5, LO7, LO8	1	Getting Started with Tableau			
30/09/25	3	Visualising Trends	LO5, LO6, LO7, LO8	2	Data Visualisation	2%		
07/10/25	4	Visualising Comparisons	LO5, LO6, LO7, LO8	3	Dashboards in Tableau	2%	1 (30%)	
14/10/25	5	Visualising Relationships	LO5, LO6, LO7, LO8	4	Exploratory Analysis			
21/10/25	6	Introduction to R	LO1, LO2, LO3, LO4	5	R Tutorial and Setup			
28/10/25	7	Review Week	Reading Week					1 (30%)
04/11/25	8	Encodings I	LO1, LO2, LO3, LO4	6	Visualisation in R	2%	2 (30%)	
11/11/25	9	Encodings II	LO1, LO3, LO5, LO6, LO7	7	Visualisation in R	2%		
18/11/25	10	Spatial Relationships I	LO3, LO5, LO6, LO7	8	Visualisation in R + Animation	2%		
25/11/25	11	Spatial Relationships II	LO3, LO5, LO6, LO7					
02/12/25	12	Visualising Text	LO5, LO6, LO7, LO8					2 (30%)
09/12/25	13	Course Summary	All					

VISUALISATION

Drowning in Data

- The Large Hadron Collider at CERN is the largest scientific instrument ever constructed
- When turned on, Particles collide in the LHC detectors approximately 1 billion times per second, generating about one petabyte of collision data per second
 - Storing only 'interesting' data keeps one petabyte a day
 - Petabyte = 1m Gb



15 Minutes of Fame

- In 2010, more than 13 million hours of video were uploaded to YouTube
- In two months (60 days) more video was uploaded than had been created in the previous six decades by the three major US television networks (ABC, CBS, NBC)
- By May 2011, more than 48 hours of video were being uploaded per minute

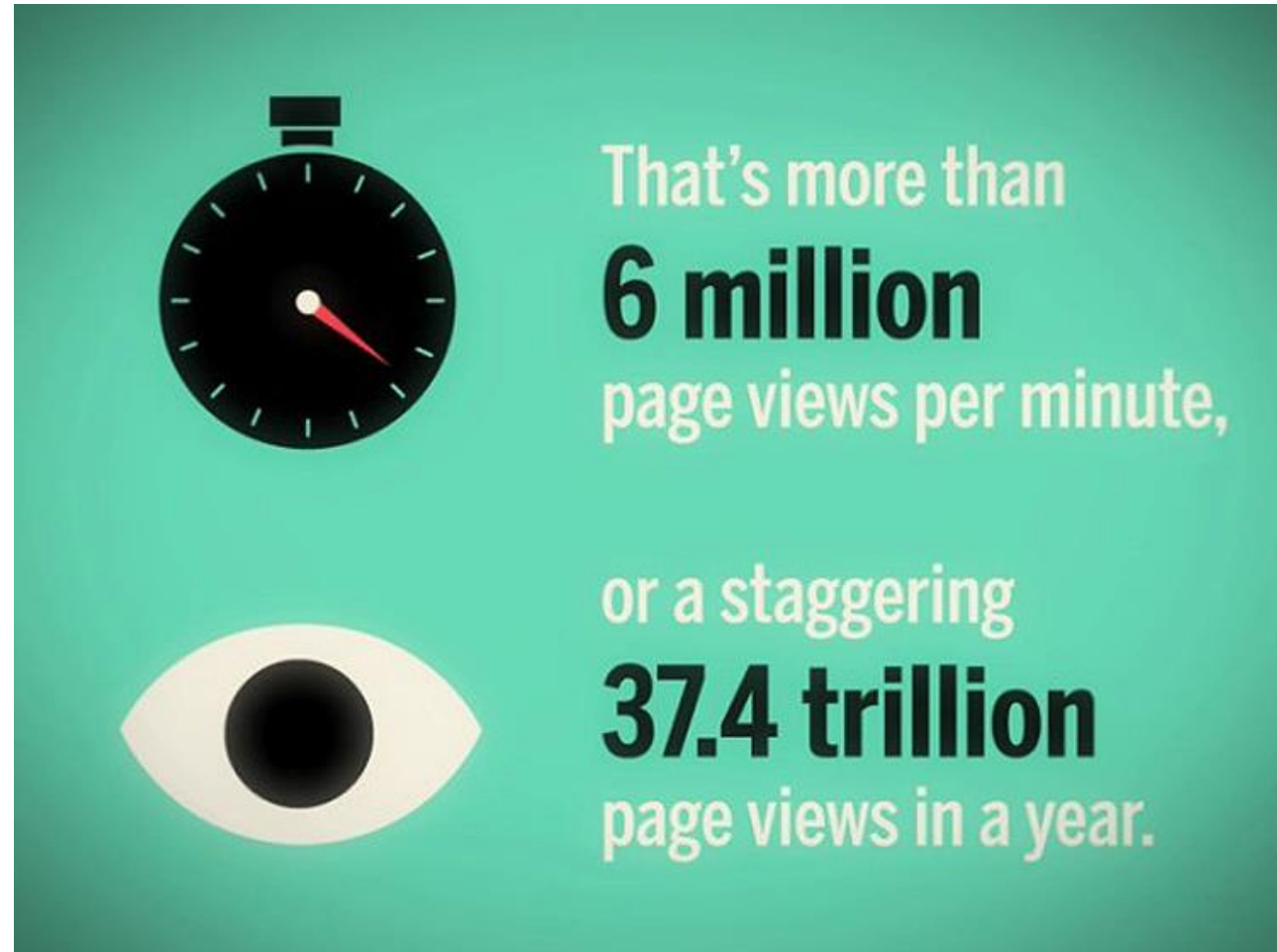


15 Minutes of Fame

- By 2019
 - Video
 - 500 hours per minute 720000 hours per day
 - 5+ billion videos watched per day
 - 5 billion videos shared to date
 - Users
 - 2 billion log-in monthly users
 - Most viewed video almost 7 billion views
 - First 24 hours 43.2 million views



State of the internet



Example: Taxi!

- I order a mytaxi from the airport
- How can this journey be logged?



Example: Taxi!

- The taxi operator can record the time the taxi was ordered, the pick-up and destination locations, the customer's details
- The phone company can record the phone call maker and receiver, the duration of the call
- The credit card processor can record the transaction amount, transaction time, transaction type
- The credit card issuer can record the billing record, the interest rate, the current available credit

Example: Taxi!

- The car maintenance service can record the fuel usage, the engine state
- The passenger can record the expense of the taxi
- The DAA can record the time the taxi drop off, the number of passengers in the car
- Dublin City Council can record the passage of the taxi across the city
- The taxi driver can record the time of the fare, the type of fare, the amount of the fare
- The taxi company can record the path taken by the driver, the duration of the trip

Questions?

- So how can we understand this data?
- So how can we learn for this data?
- How can we make this data accessible?
- How can we gain insights into this data?

What is Visualisation?

- From OED:
 1. The action or fact of visualizing; the power or process of forming a mental picture or vision of something not actually present to the sight; a picture thus formed.
 2. The action or process of rendering visible.

What is Visualisation?

- Transformation of the symbolic into the geometric”
 - [McCormick et al. 1987]
- “... finding the artificial memory that best supports our natural means of perception.”
 - [Bertin 1967]
- “The use of computer-generated, interactive, visual representations of data to amplify cognition.”
 - [Card, Mackinlay, & Shneiderman 1999]

Why Visualisation?

- Answer questions (or discover them)
- Make decisions
- See data in context
- Expand memory
- Support graphical calculation
- Find patterns
- Present argument or tell a story
- Inspire

Why Visualisation?

Well-designed visual representations can replace cognitive calculations with simple perceptual inferences and improve comprehension, memory, and decision making

By making data more accessible and appealing, visual representations may also help engage more diverse audiences in exploration and analysis

The challenge is to create effective and engaging visualizations that are appropriate to the data

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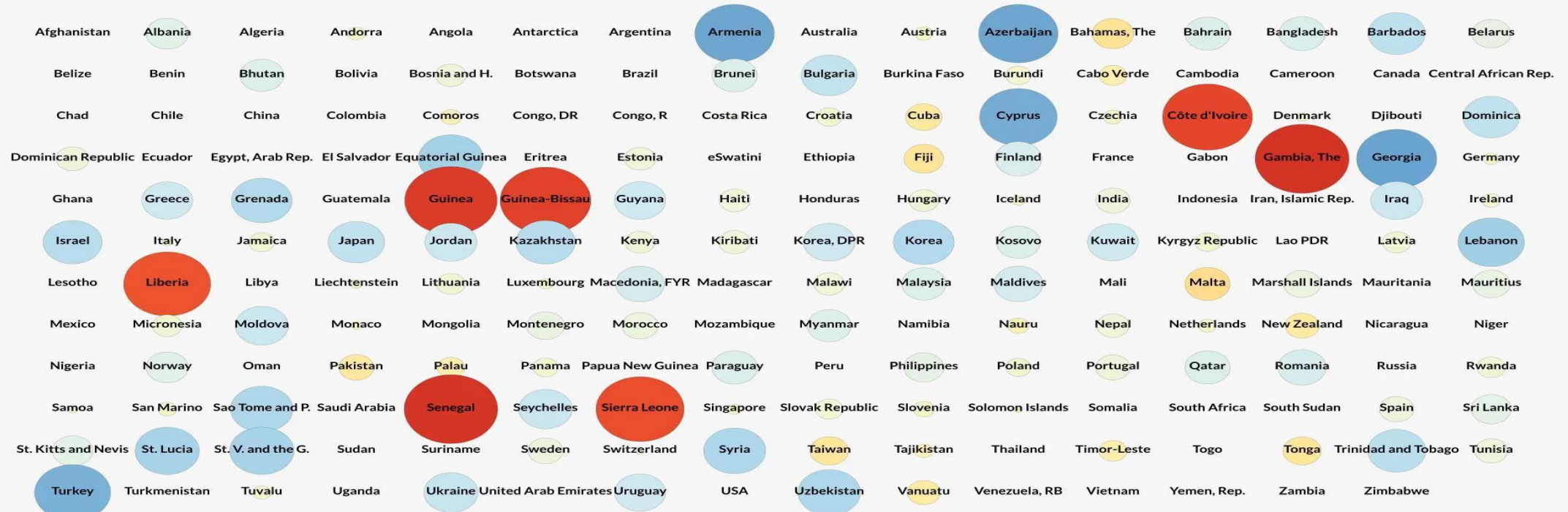
Functions of Visualisations

- **Record:** store information
 - Photographs, blueprints, ...
- **Analyse:** support reasoning about information
 - Process and calculate
 - Reason about data
 - Feedback and interaction
- **Communicate:** convey information to others
 - Share and persuade
 - Collaborate and revise
 - Emphasize important aspects of data

What is a Visualisation?

Temperature Anomalies by Country Years 1880 - 2017

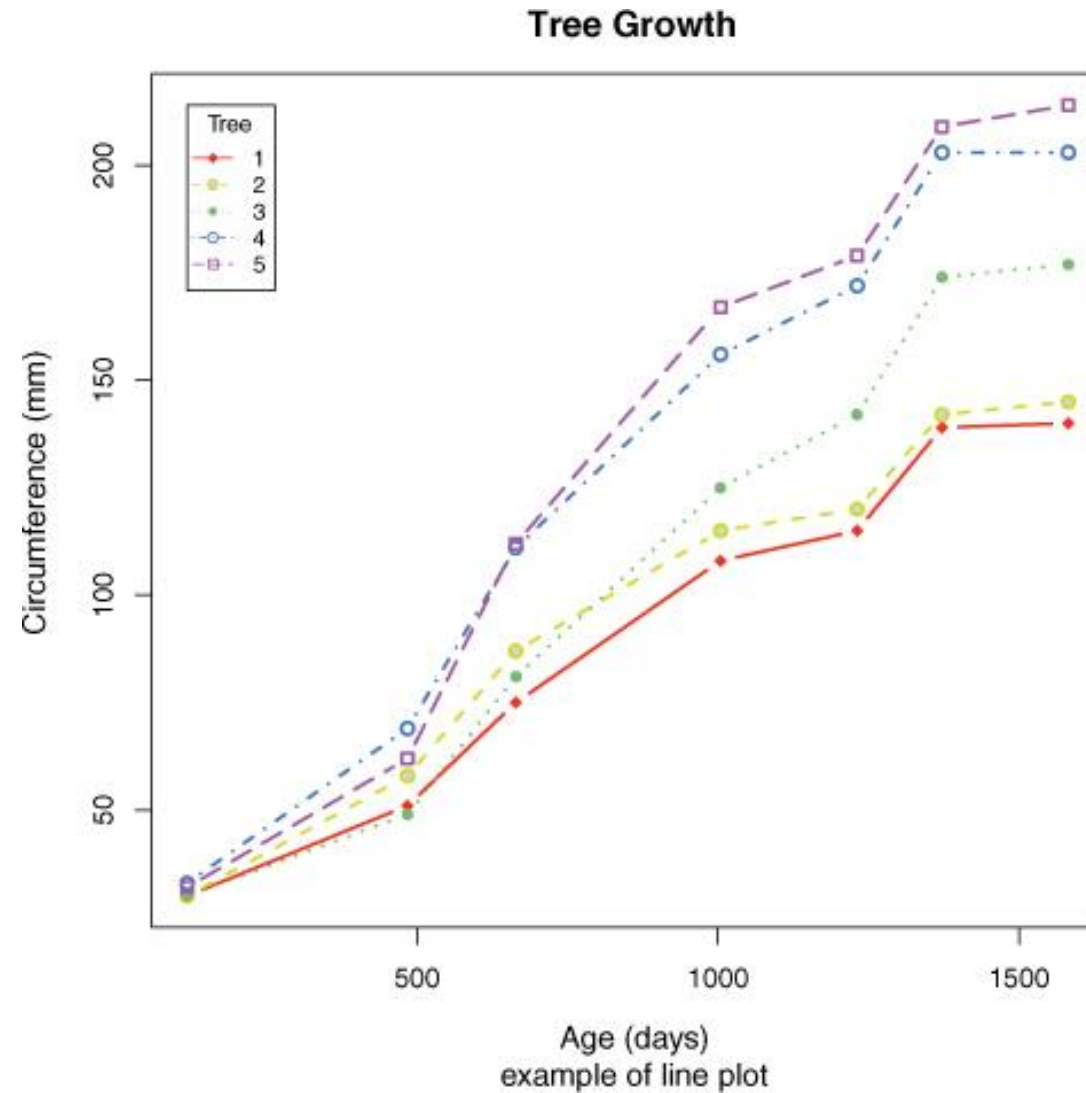
1880



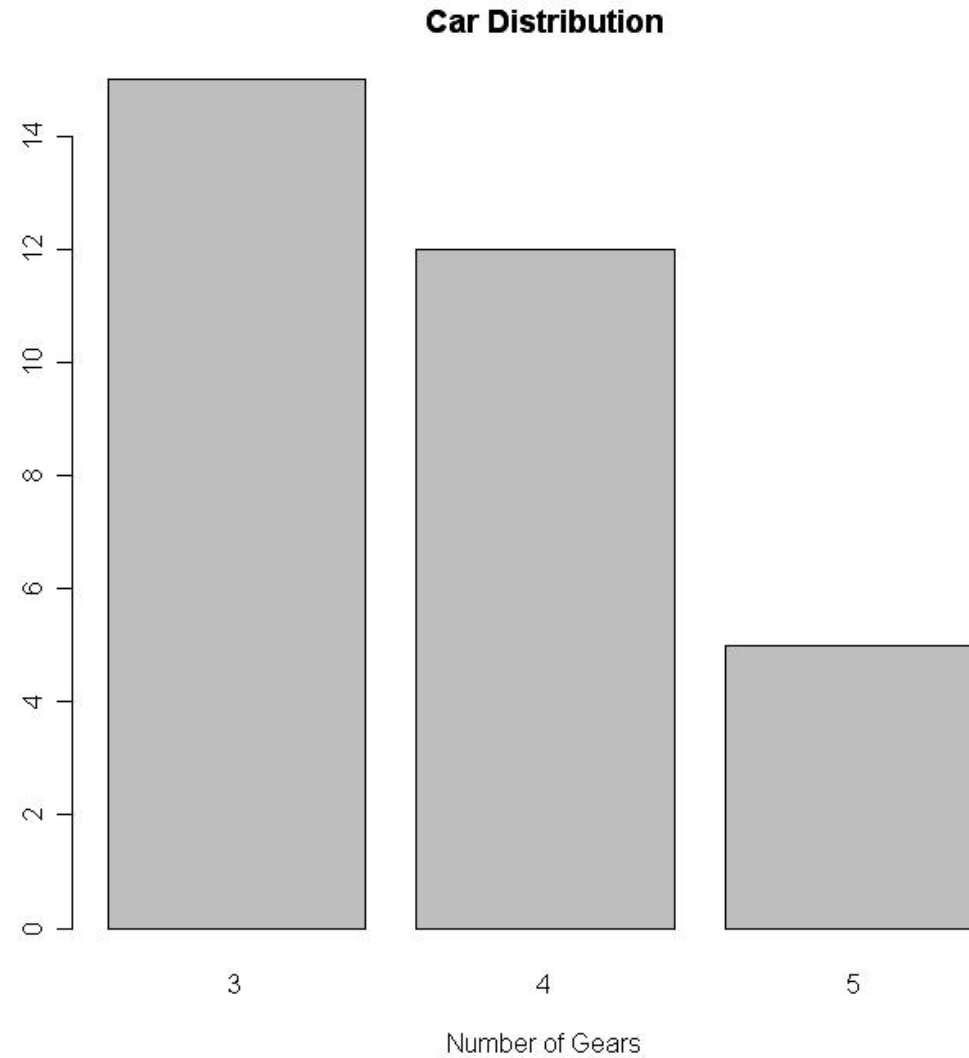
Data Source:
NASA GISS, GISTEMP Land-Ocean Temperature Index (LOTI), ERSSTv5, 1200km smoothing
<https://data.giss.nasa.gov/gistemp/>
Average of monthly temperature anomalies. GISTEMP base period 1951–1980.

Video license: CC-BY-4.0
Antti Lipponen (@anttilip)

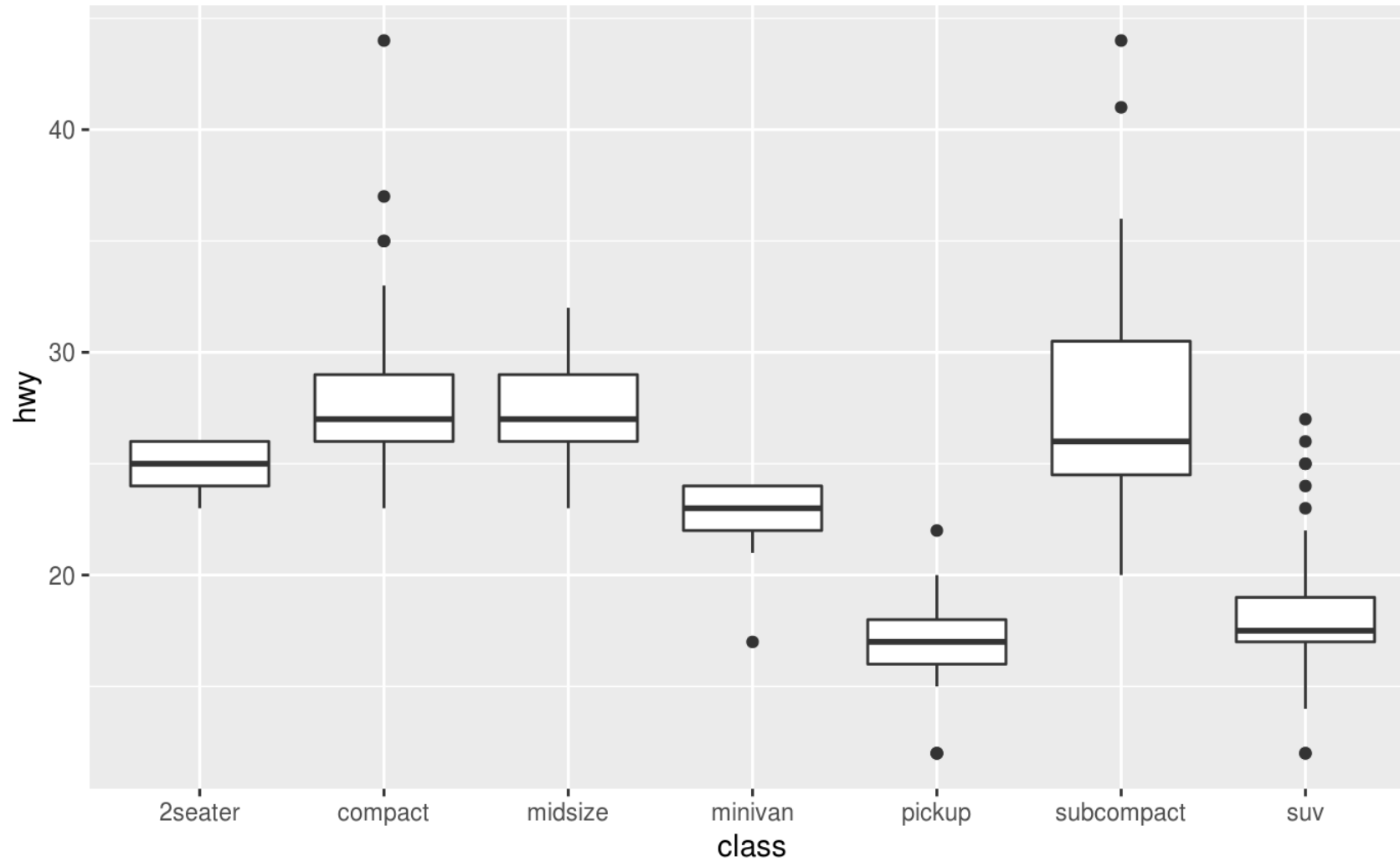
Trends Over Time



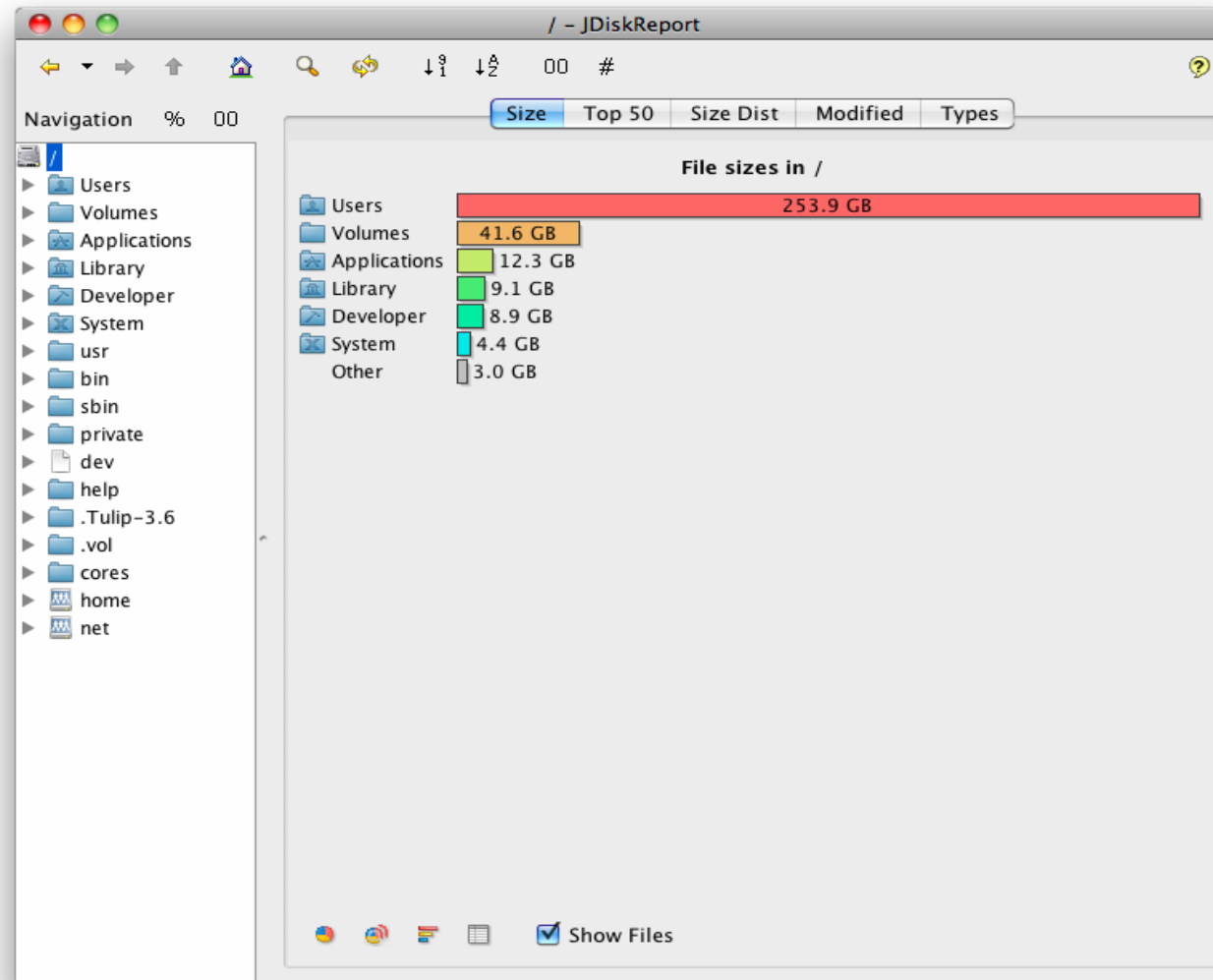
Comparisons



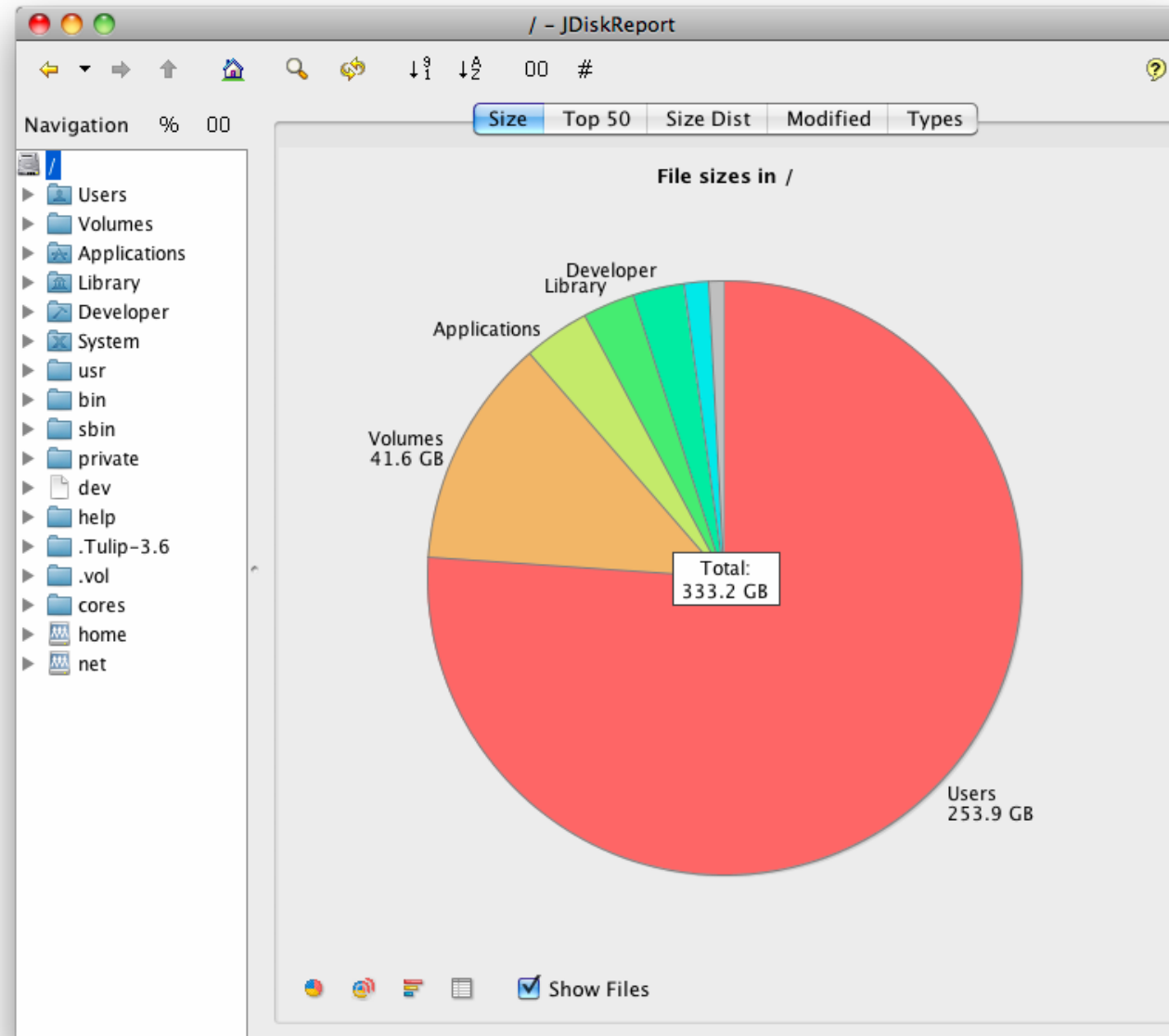
Box Plots



J Disk Report



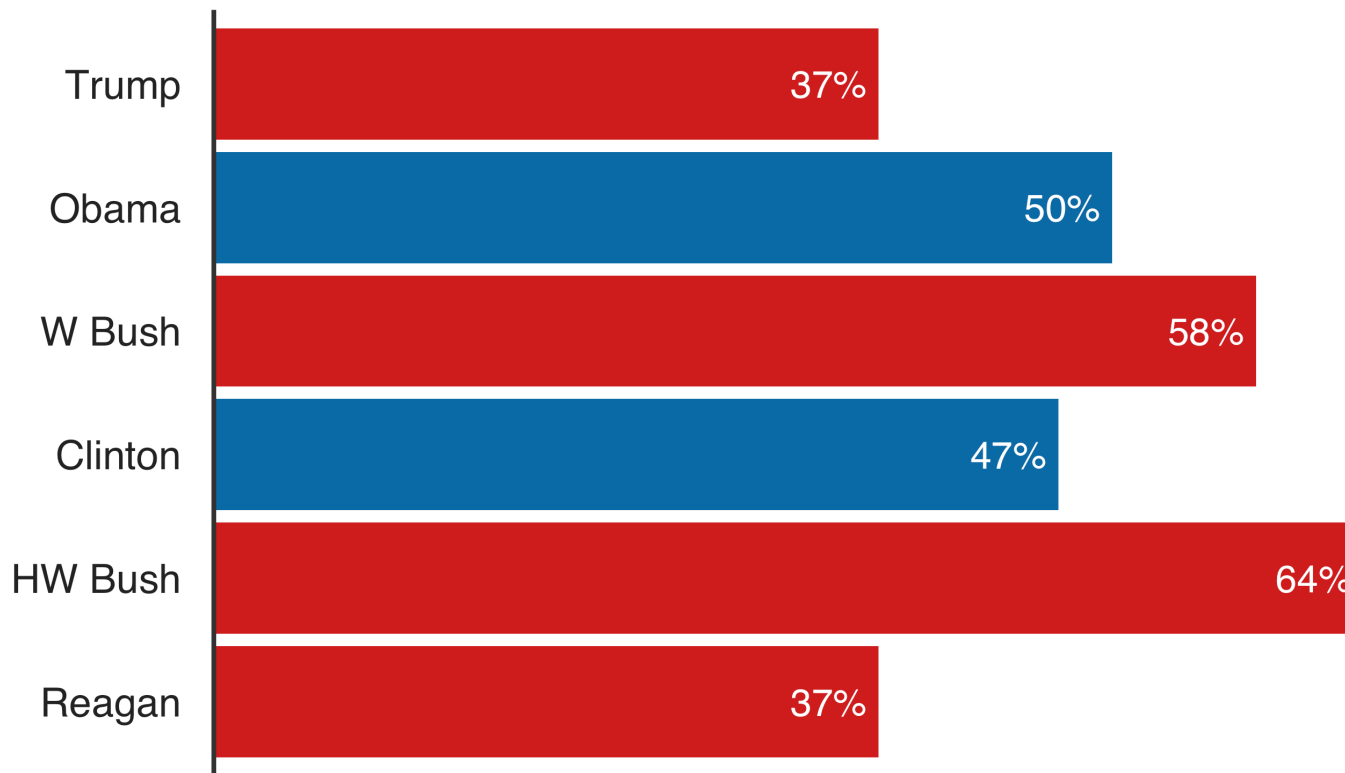
J Disk Report



Voting and Political Data

How does Trump's support compare?

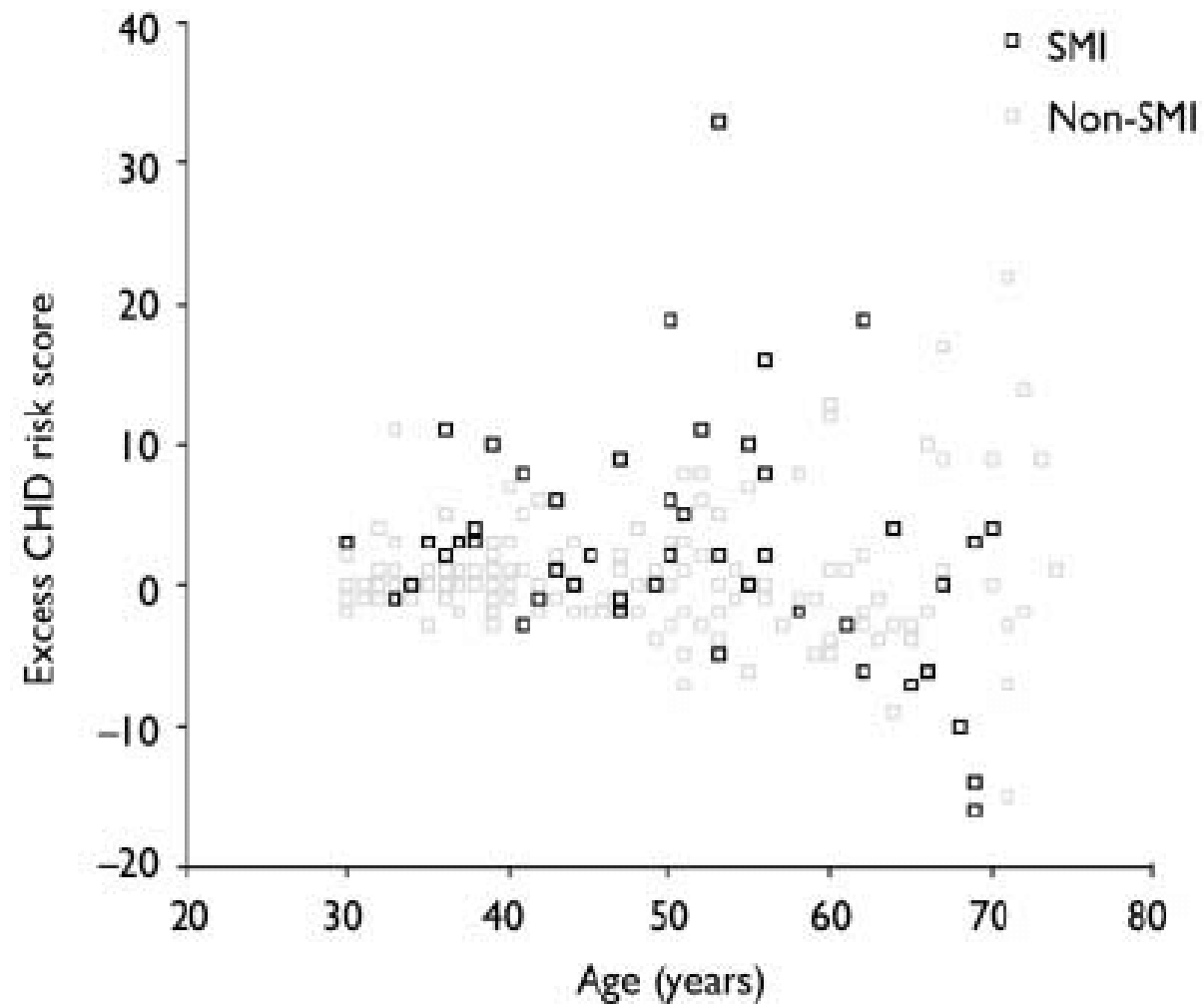
Presidential job approval rating after two years in office



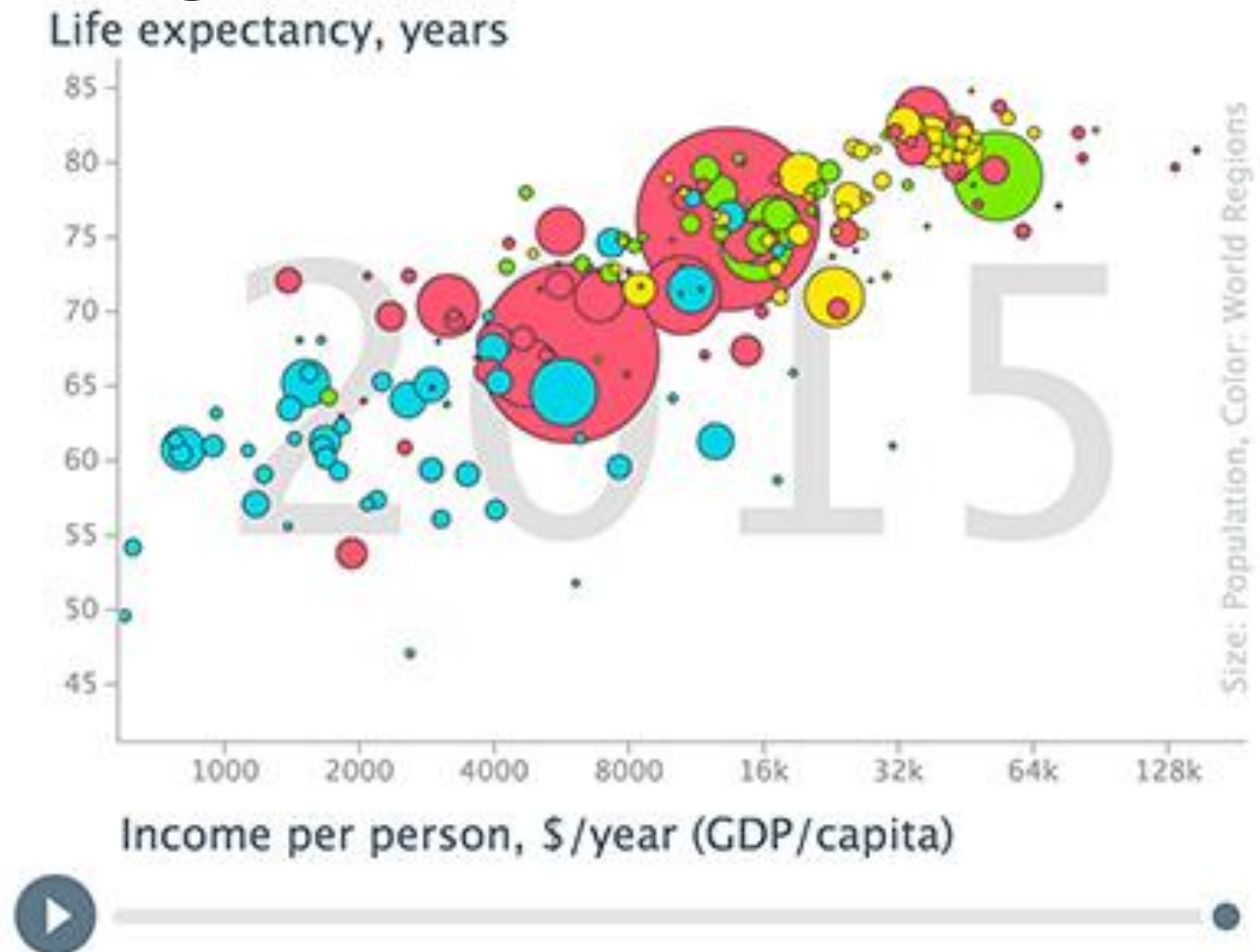
Source: Gallup



Relationships

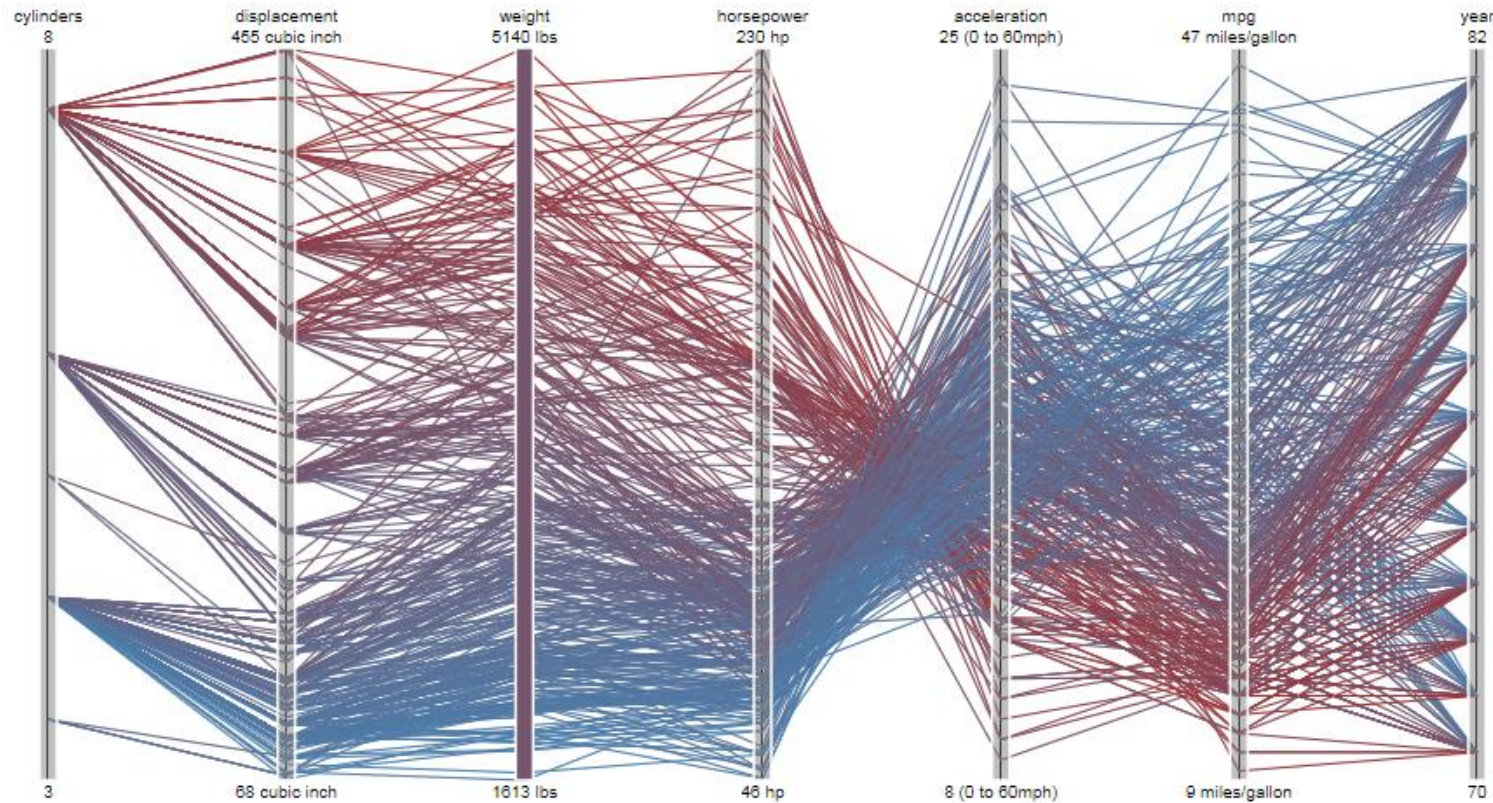


Hans Rosling



Parallel Coordinates

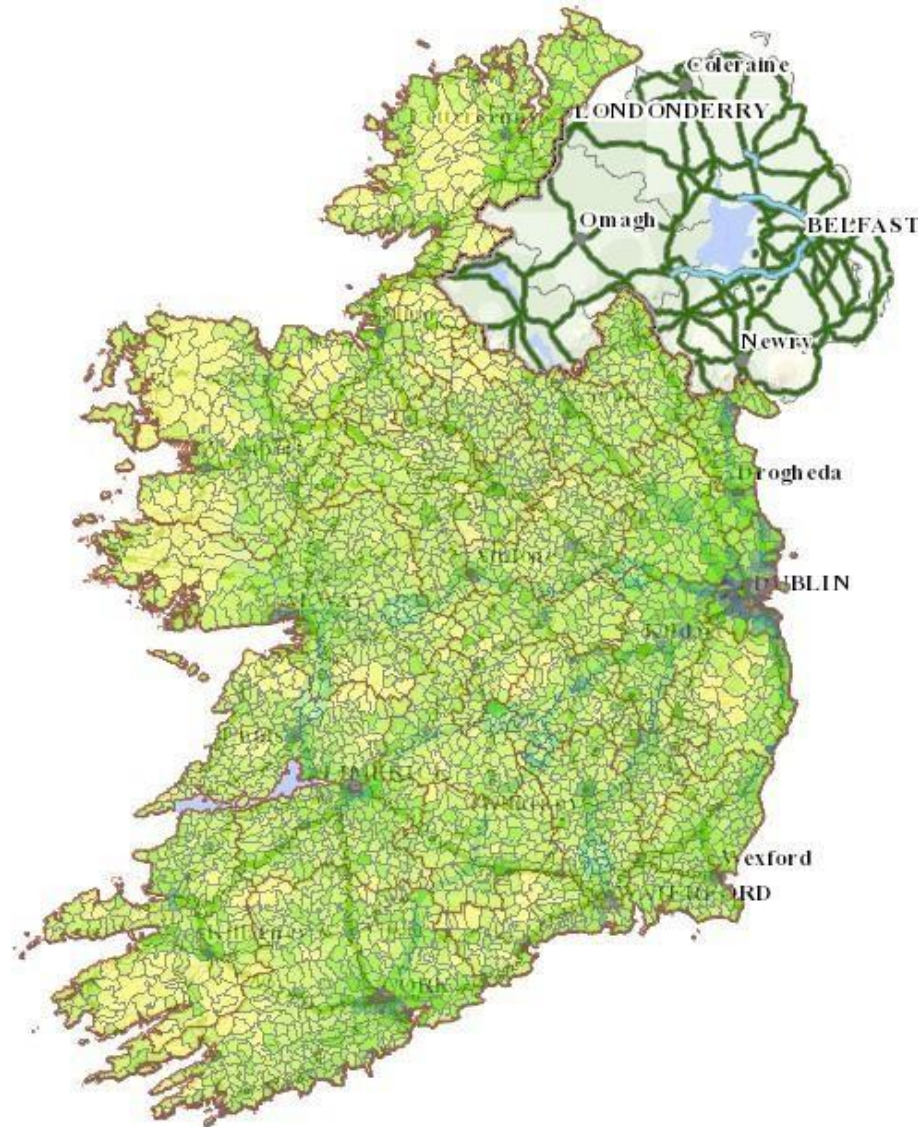
Parallel Coordinates of Automobile Data



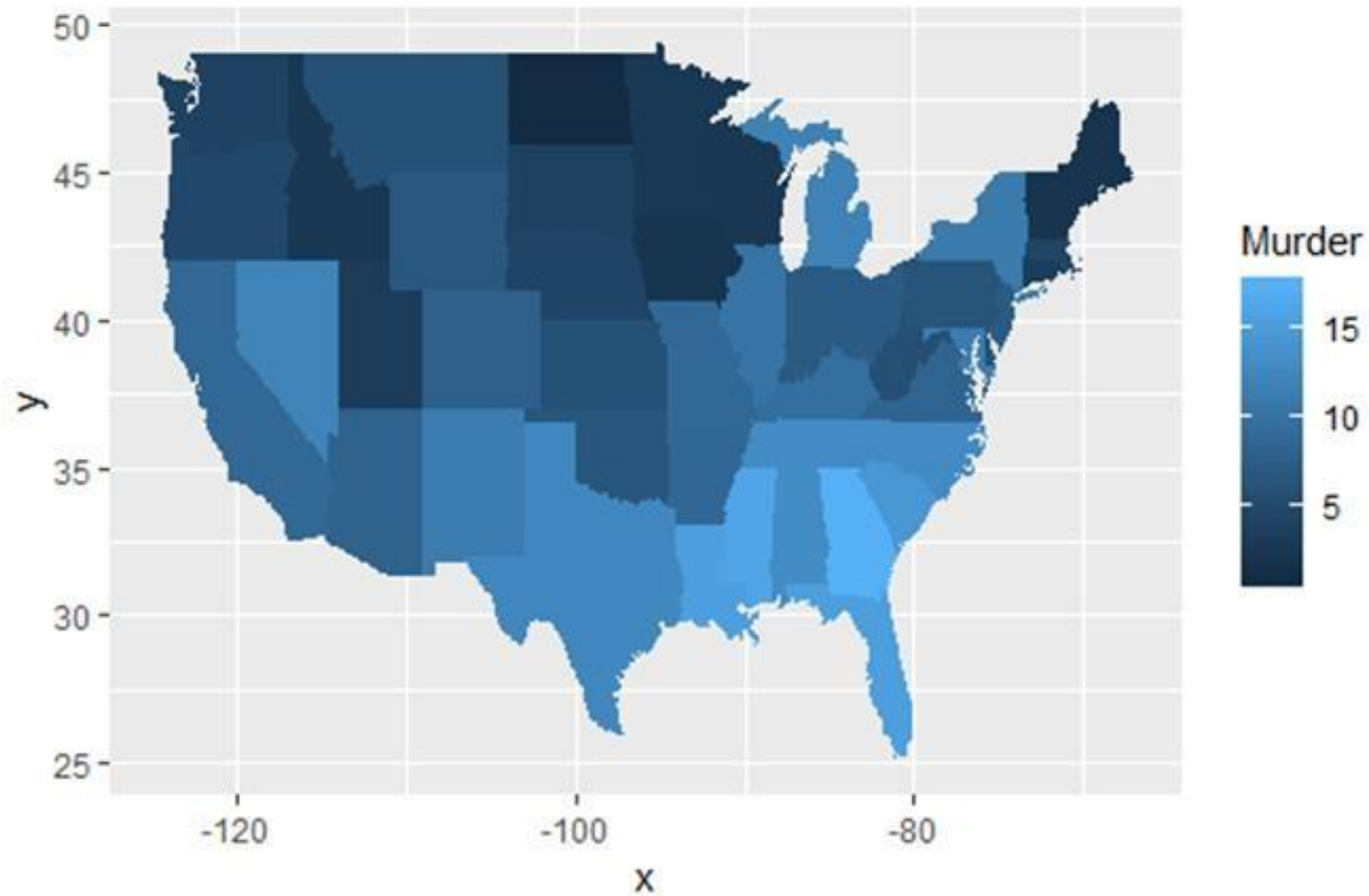
A database of cars is plotted in seven coordinate dimensions; each path represents one car. Drag and resize the coordinate selection sliders to filter the cars in any dimension.

Source: GGobi

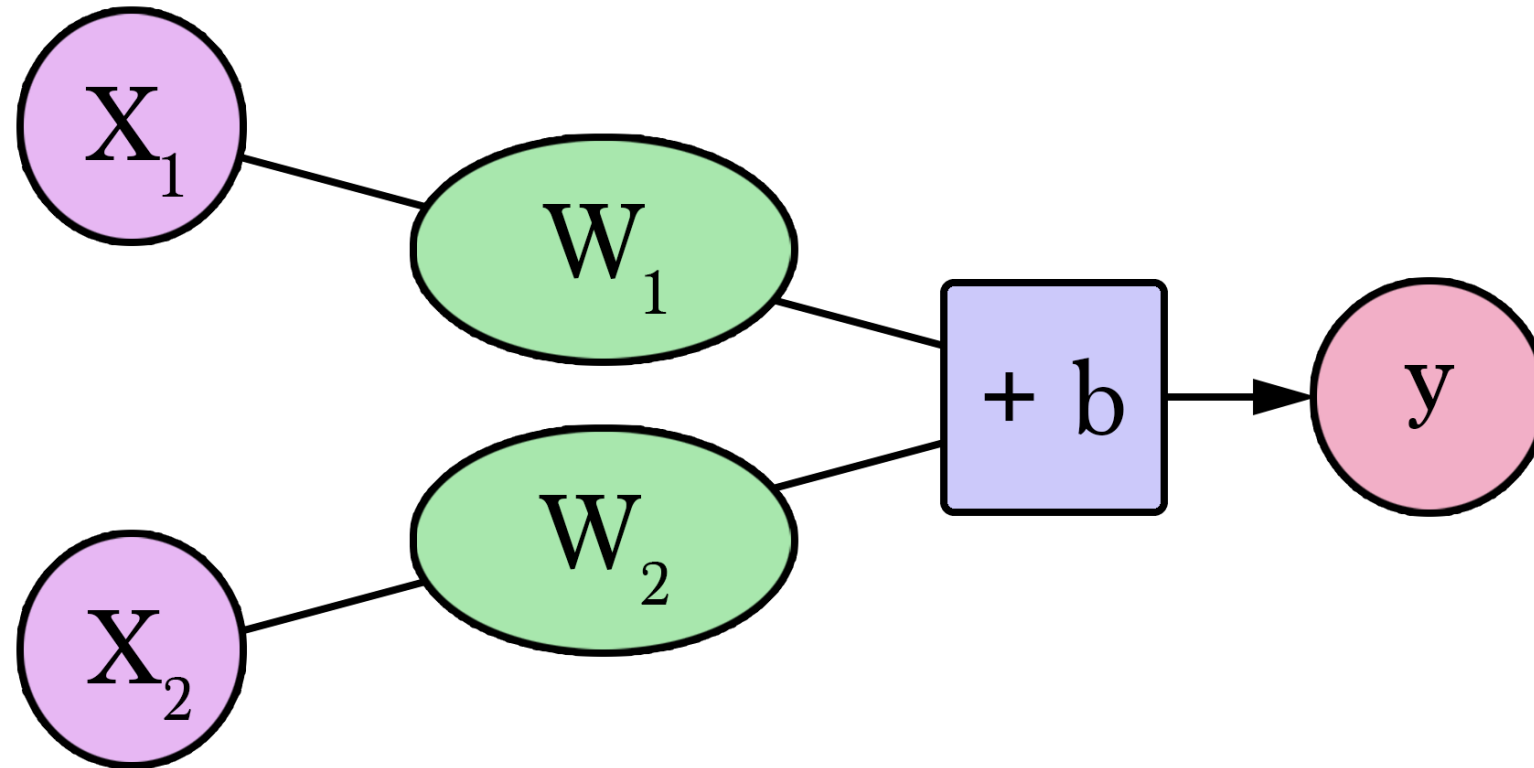
Maps



Maps – Murder rates in USA

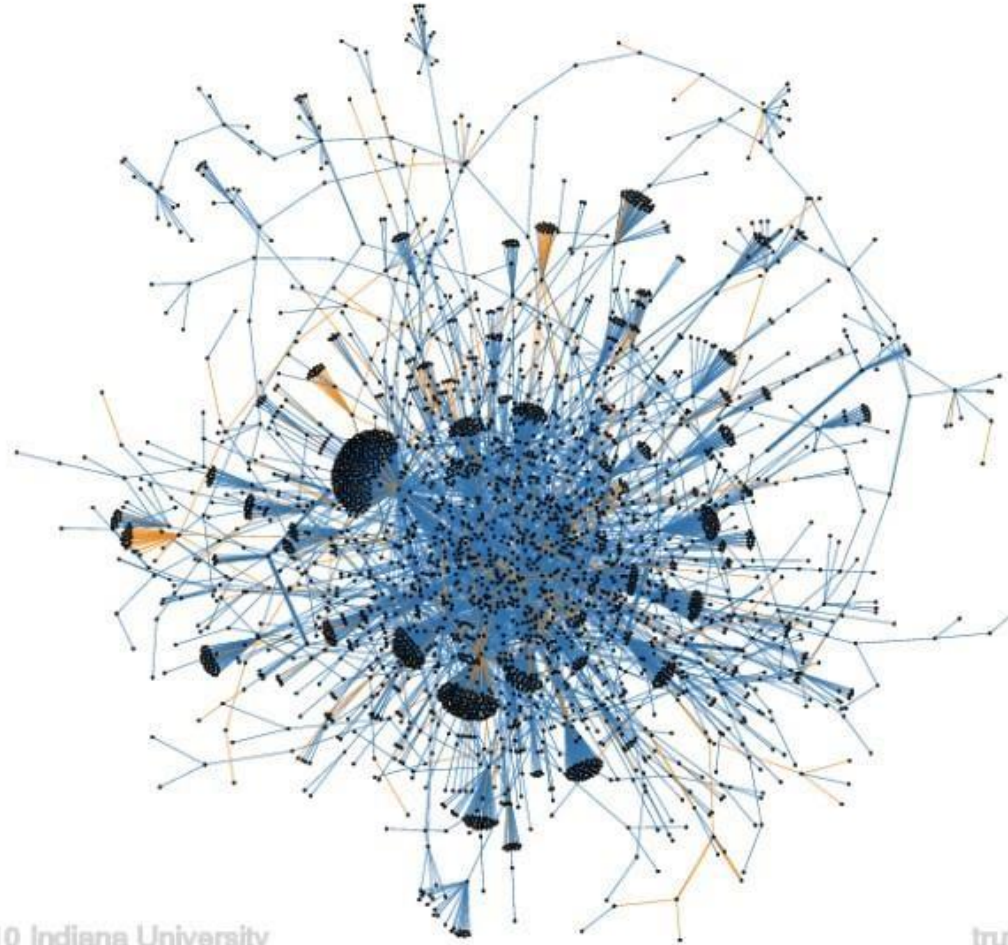


Networks



Truthy

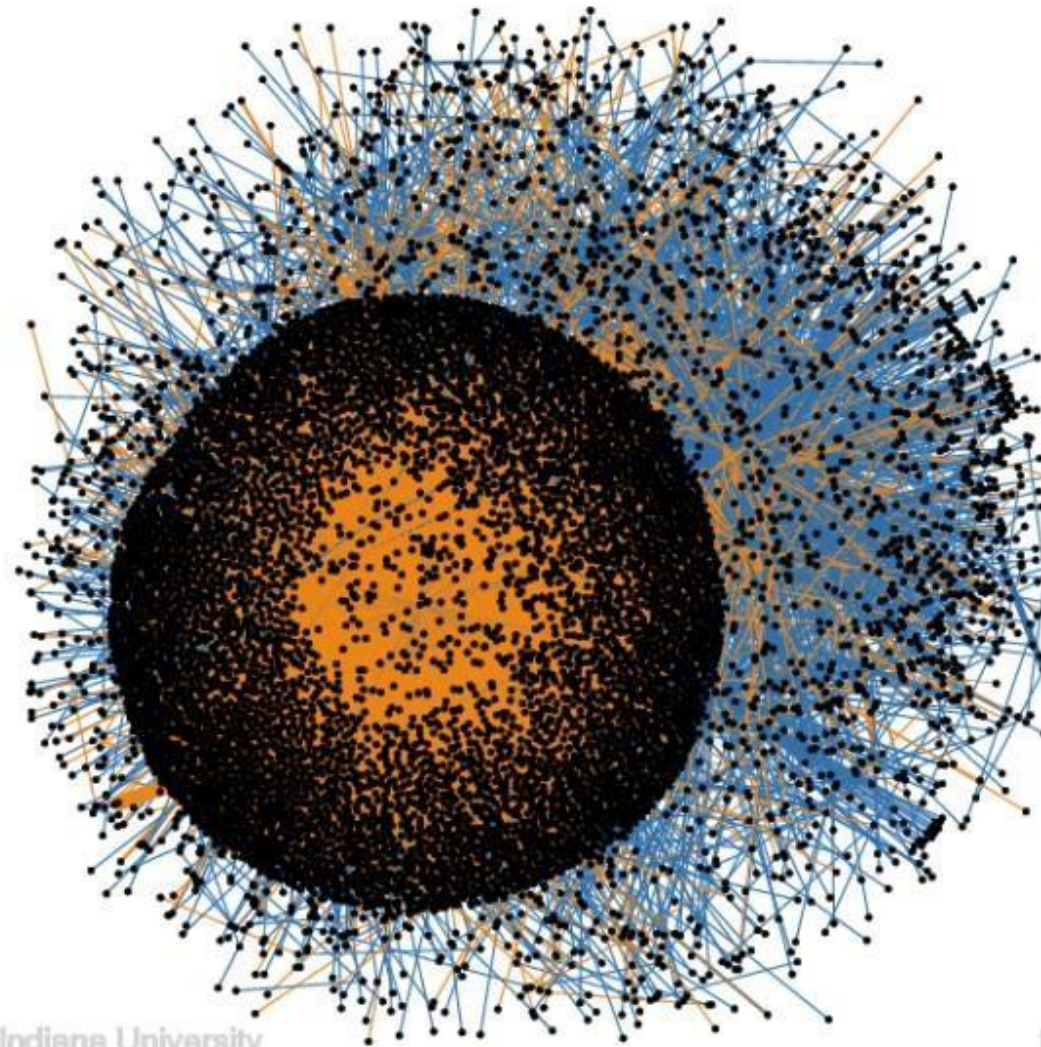
How #occupy
moved through
Twitter



Copyright 2010 Indiana University

truthy.indiana.edu

Truthy



How
@barackobama
moved through
Twitter

© Indiana University

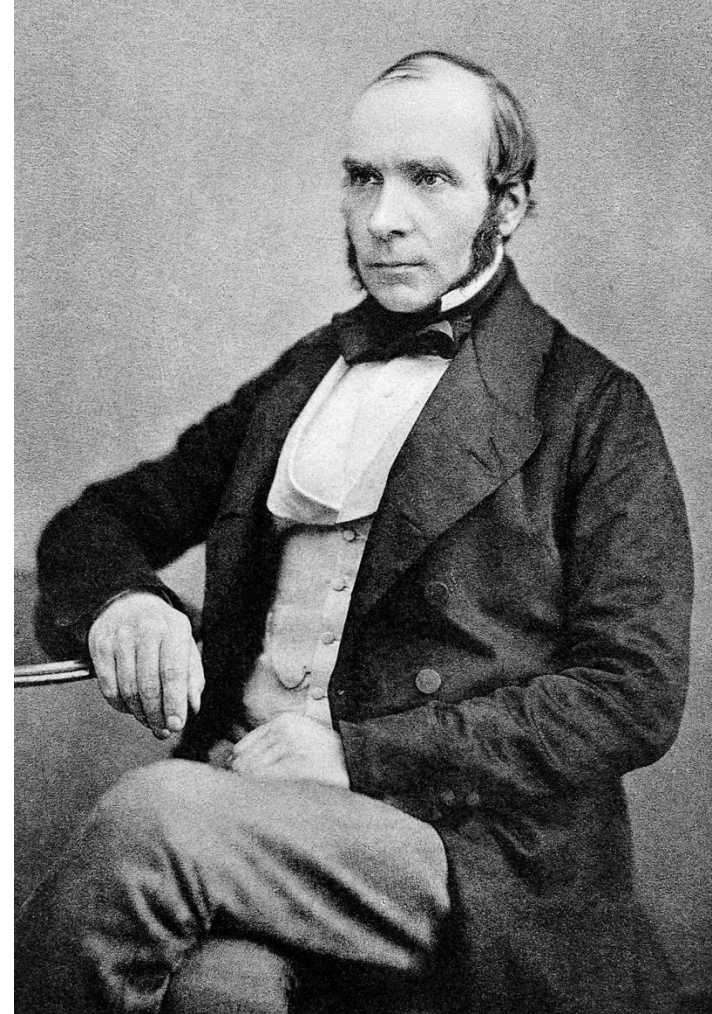
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Health InfoScape

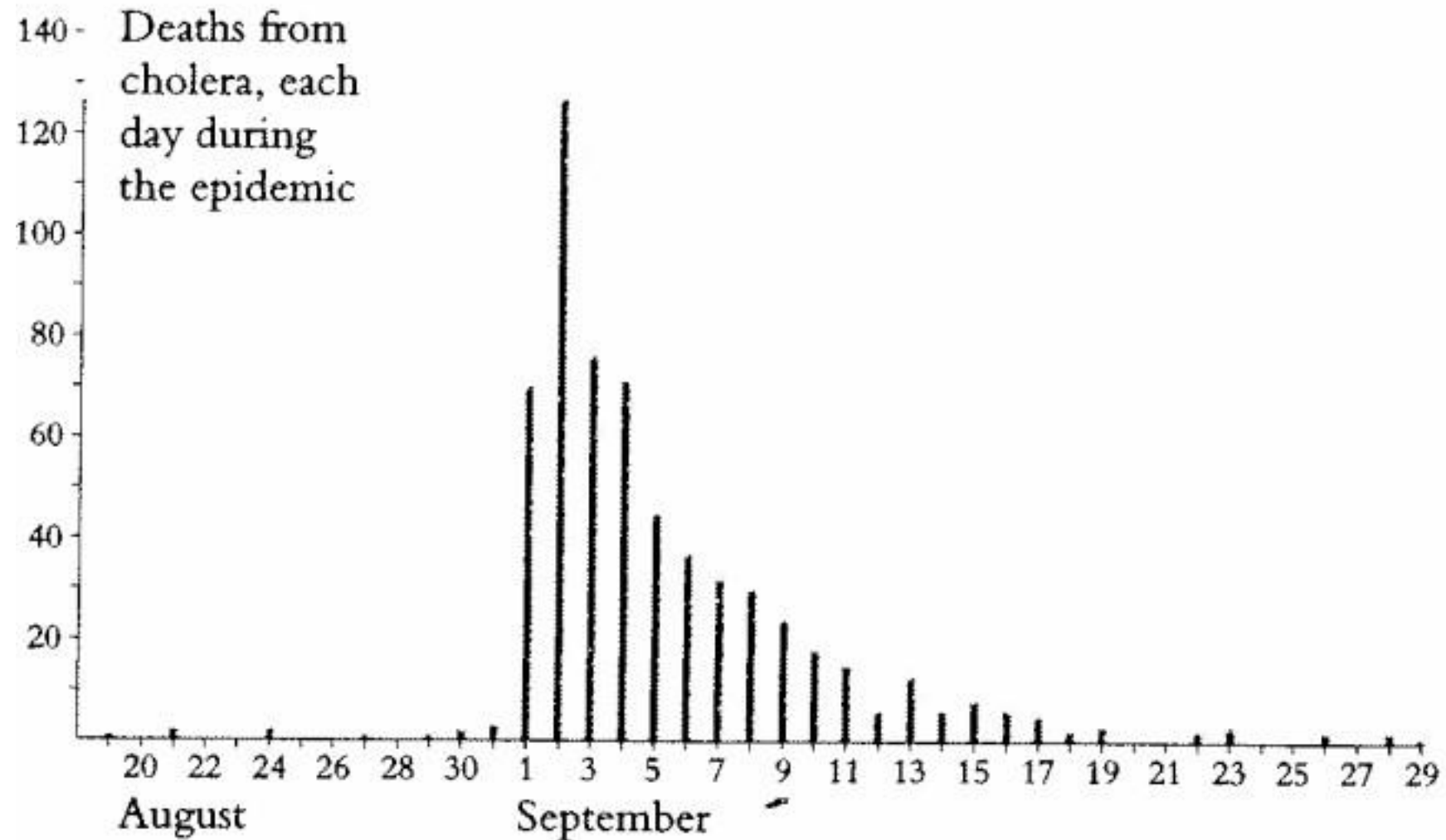


1854 London Cholera Epidemic

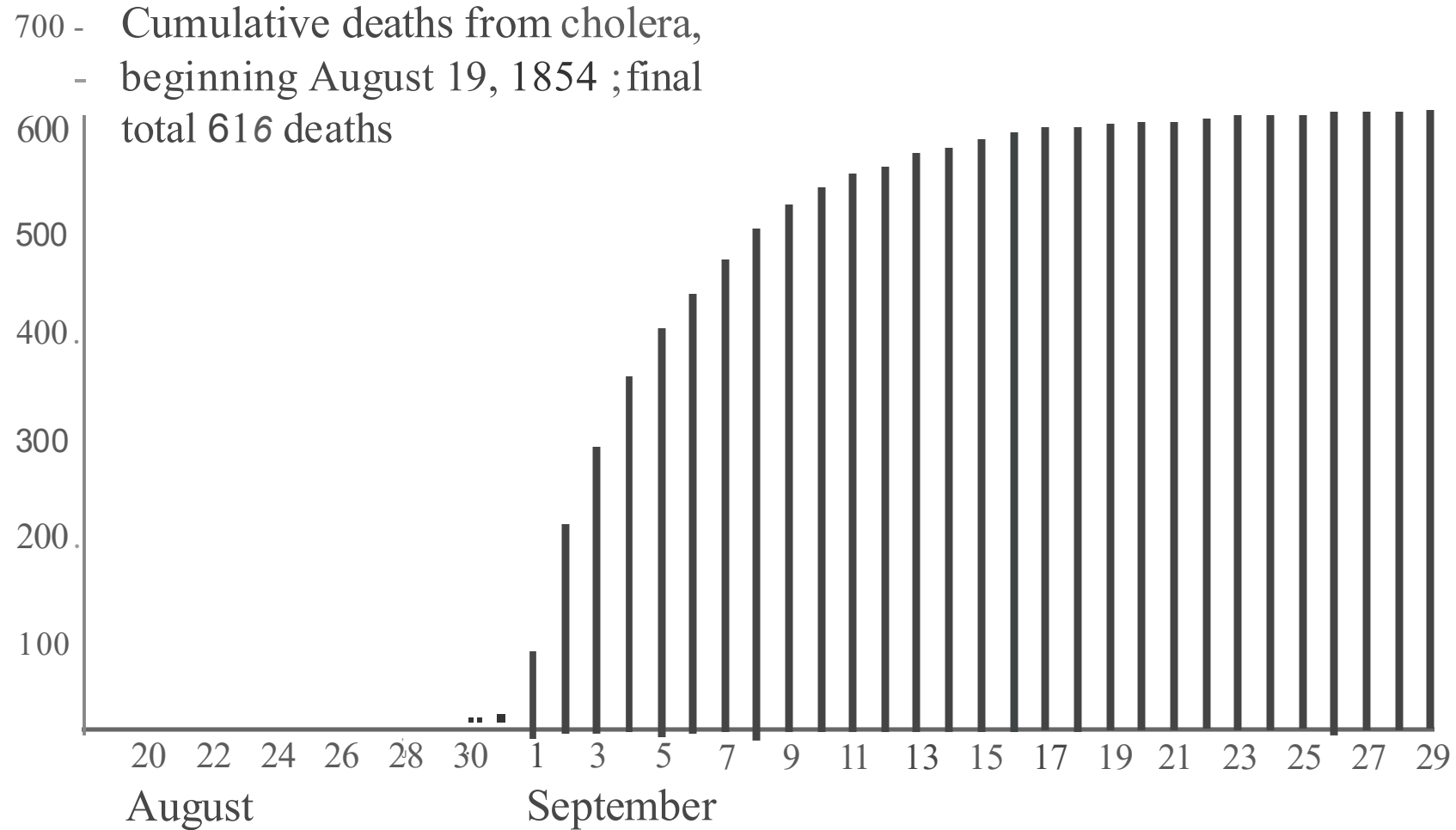
- On the evening of August 31, 1854 Cholera broke out in the Broad Street area of central London
- John Snow, a London doctor, was asked to investigate
- Visualisation was key to his solution

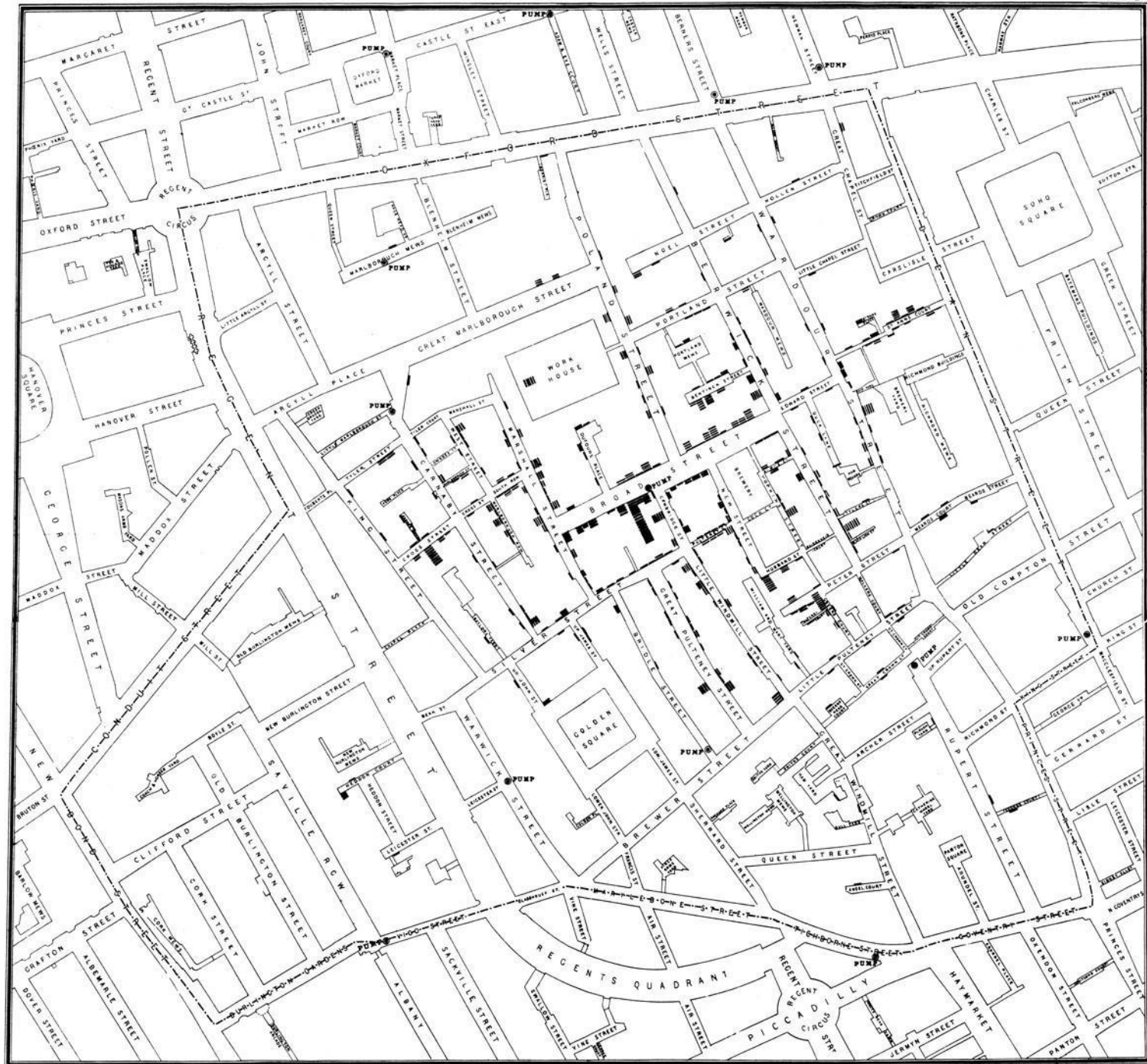


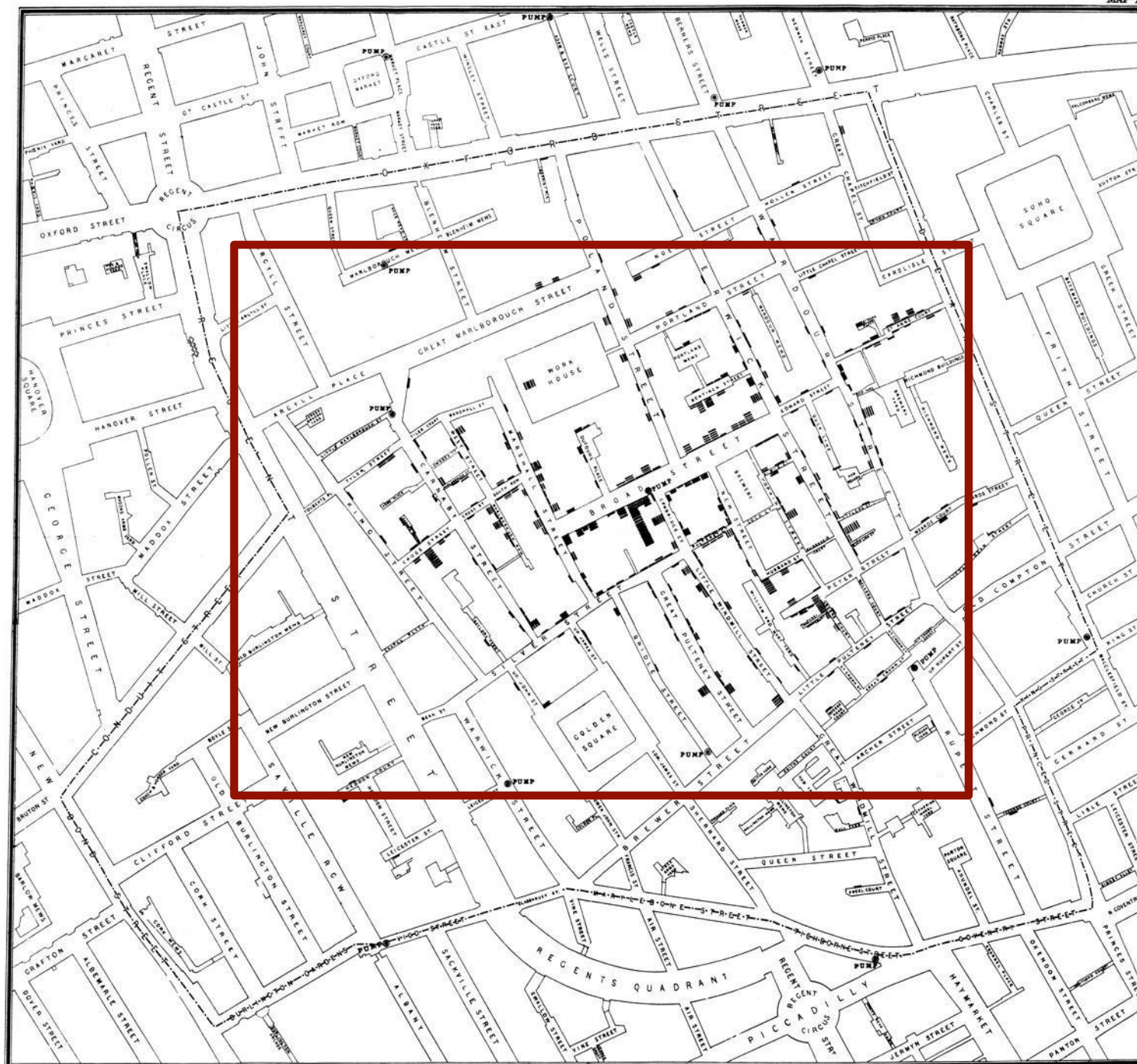
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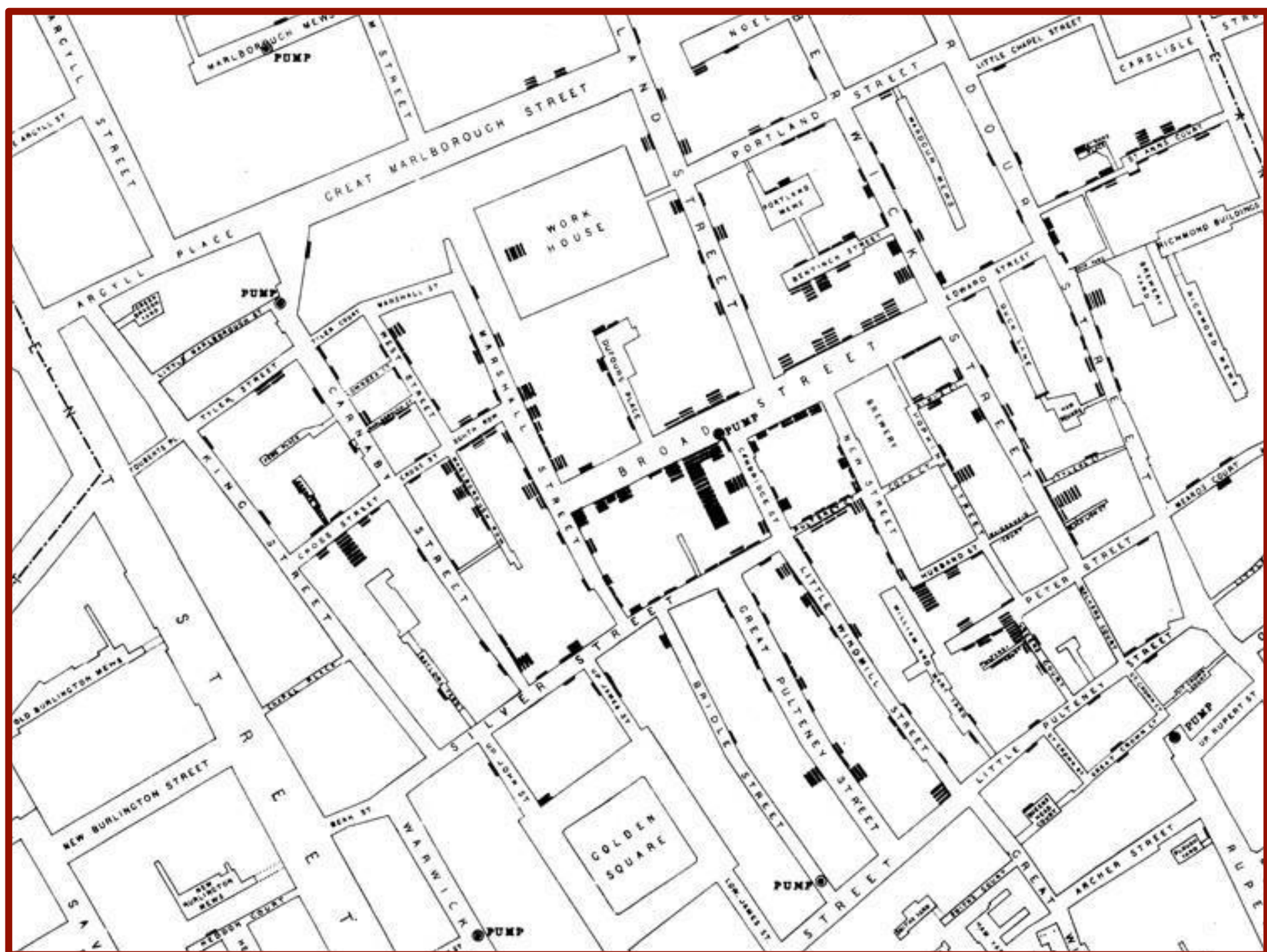


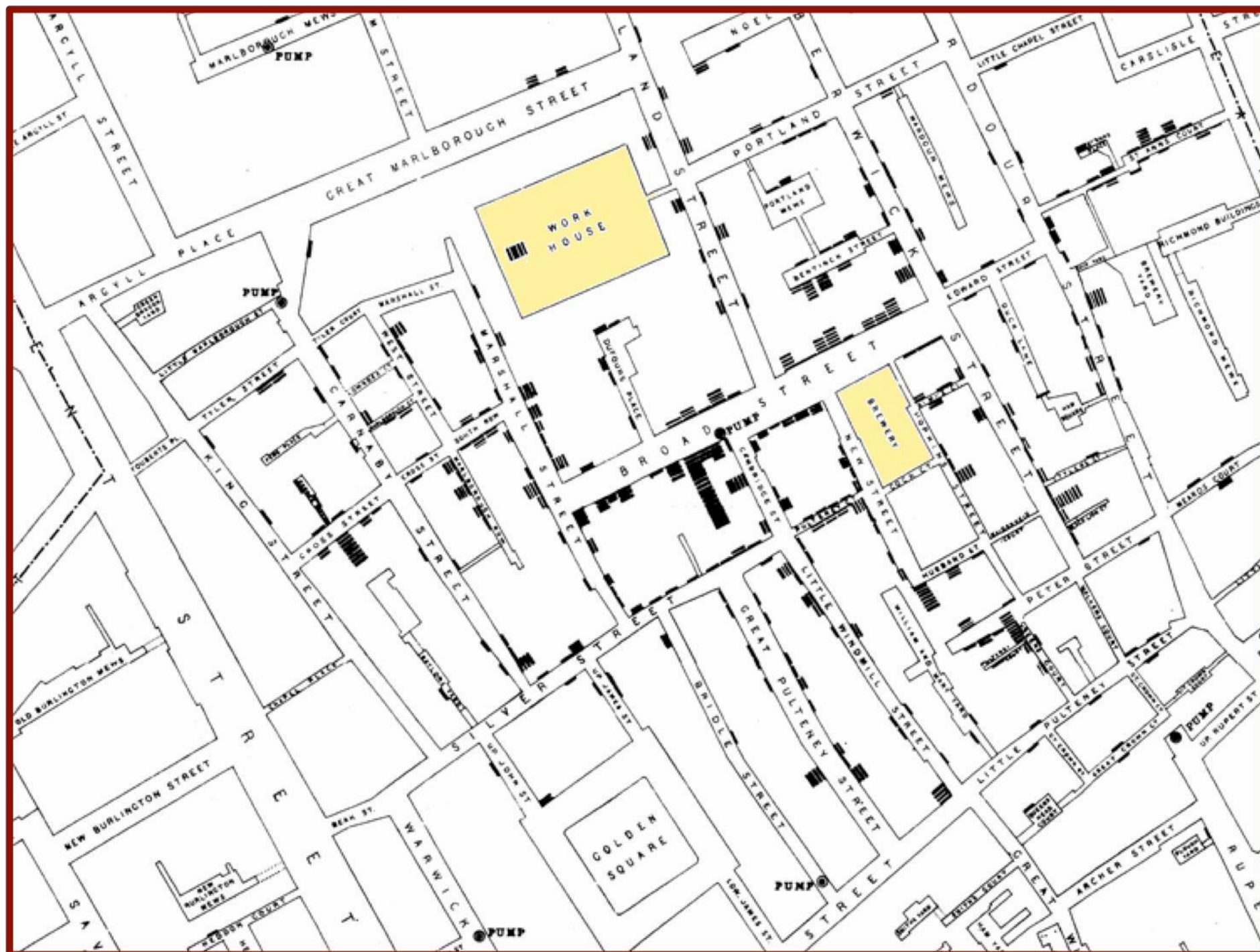
1854 London Cholera Epidemic





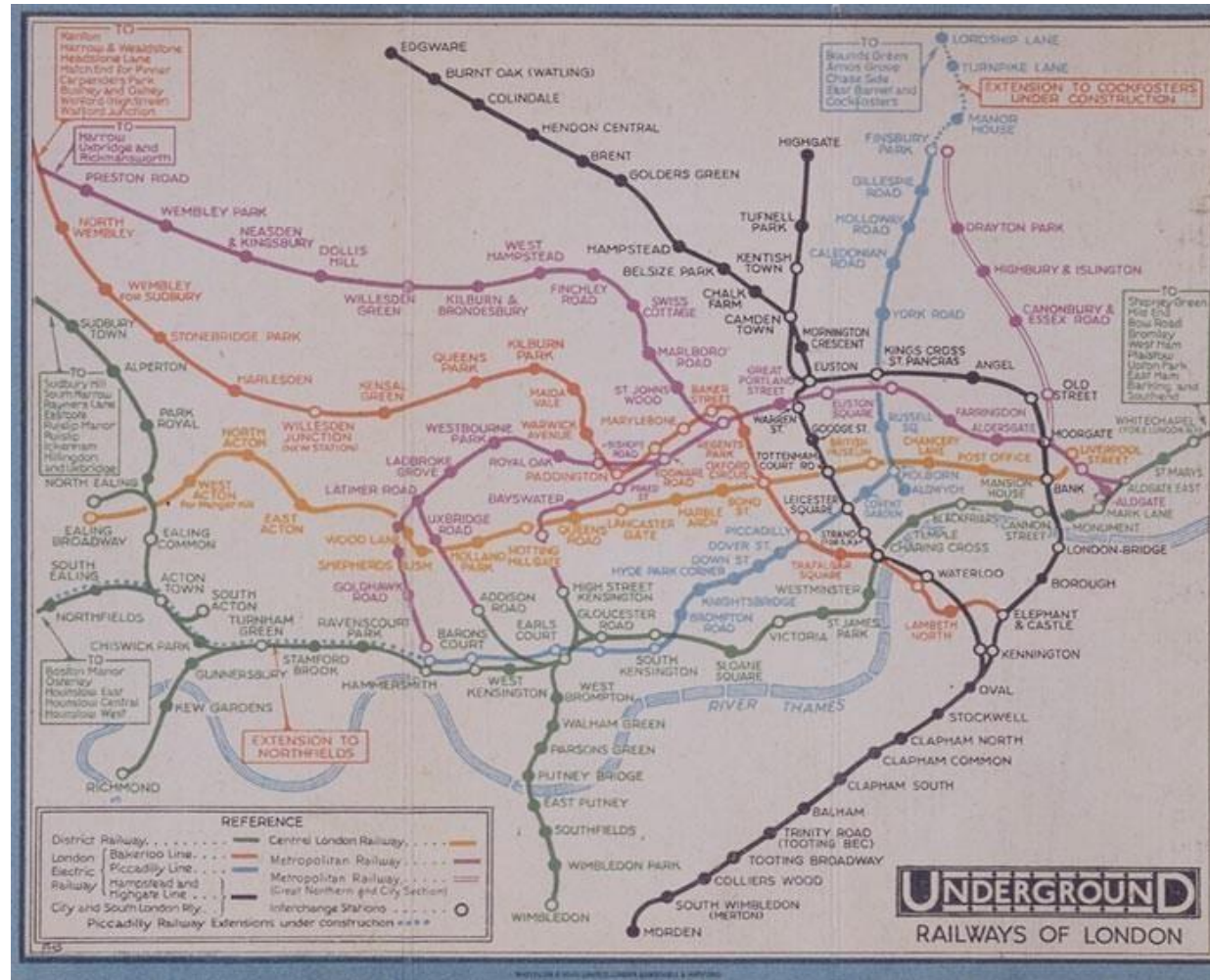








London Underground Map



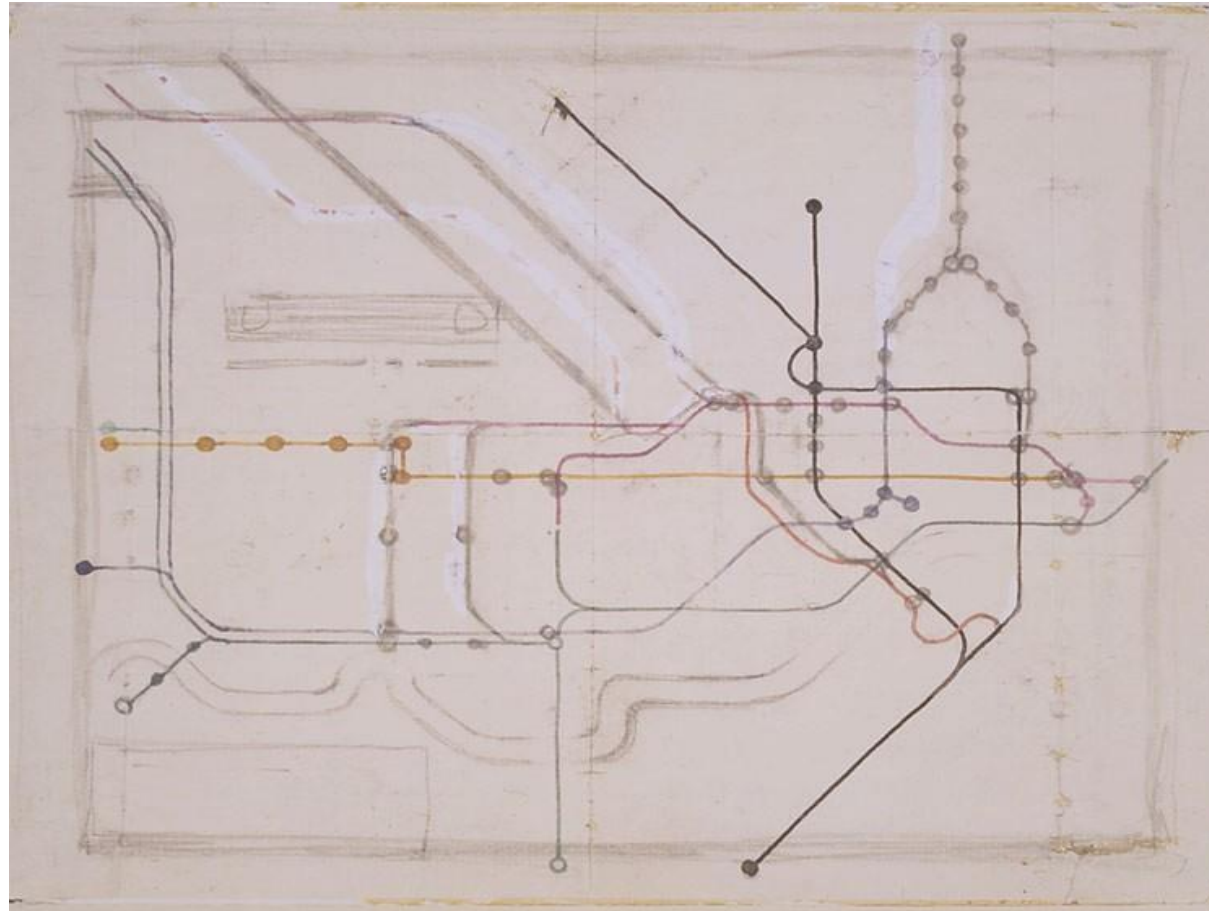
London Underground map circa 1930

London Underground Map



- Draughtsman Harry Beck:
- *"If you're going underground, why do you need bother about geography? ... Connections are the thing"*
- Designed a new map

London Underground Map



An early sketch
of Harry Beck's
new map

London Underground Map



Beck's London Underground map circa 1933

Visualisation?

- Well-designed visual representations can replace cognitive calculations with **simple perceptual inferences** and **improve comprehension, memory, and decision making**
- By making data more **accessible** and **appealing**, visual representations may also help engage more diverse audiences in exploration and analysis
- The challenge is to create **effective and engaging visualizations** that are **appropriate** to the data

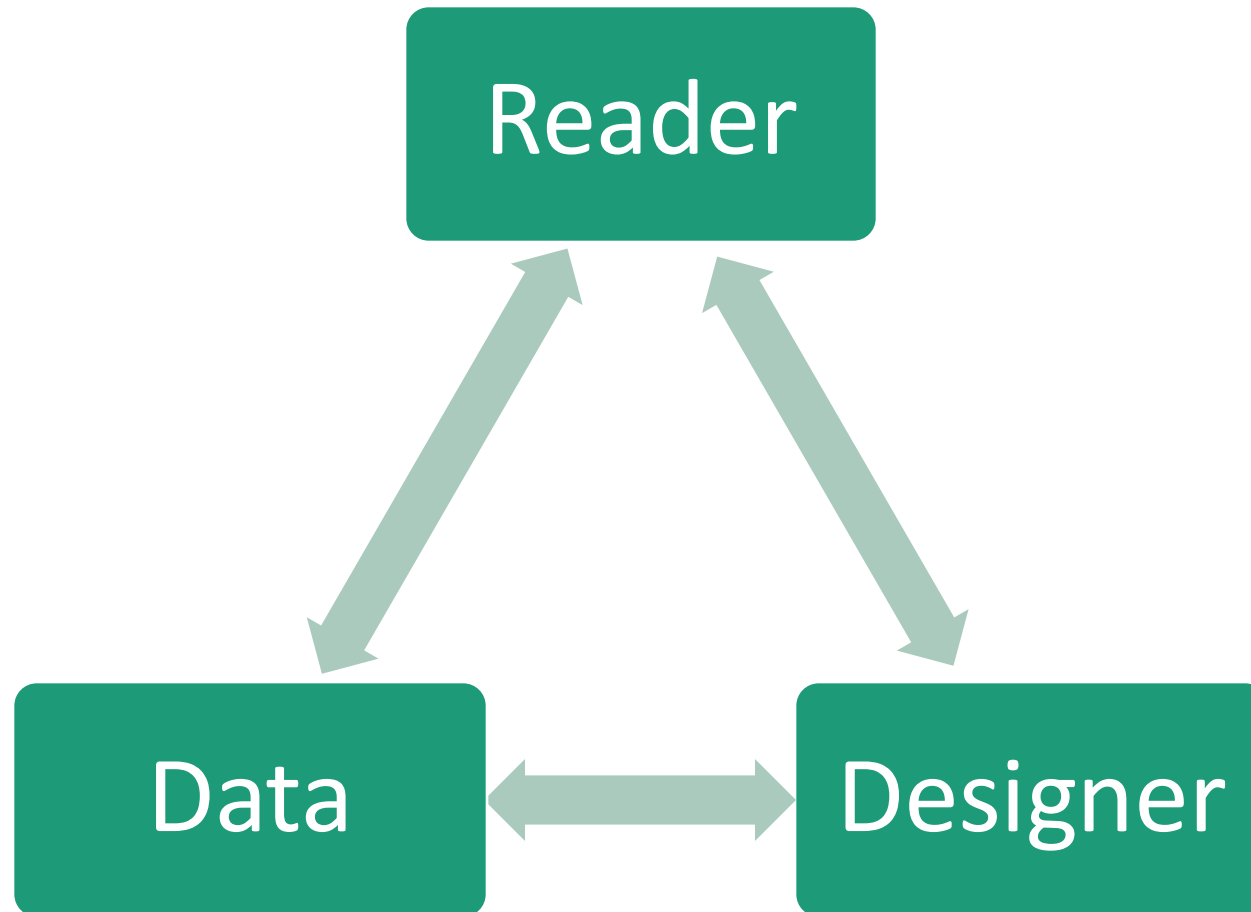
Visualisation?

*The challenge is to create **effective** and **engaging** visualizations that are appropriate to the data.*

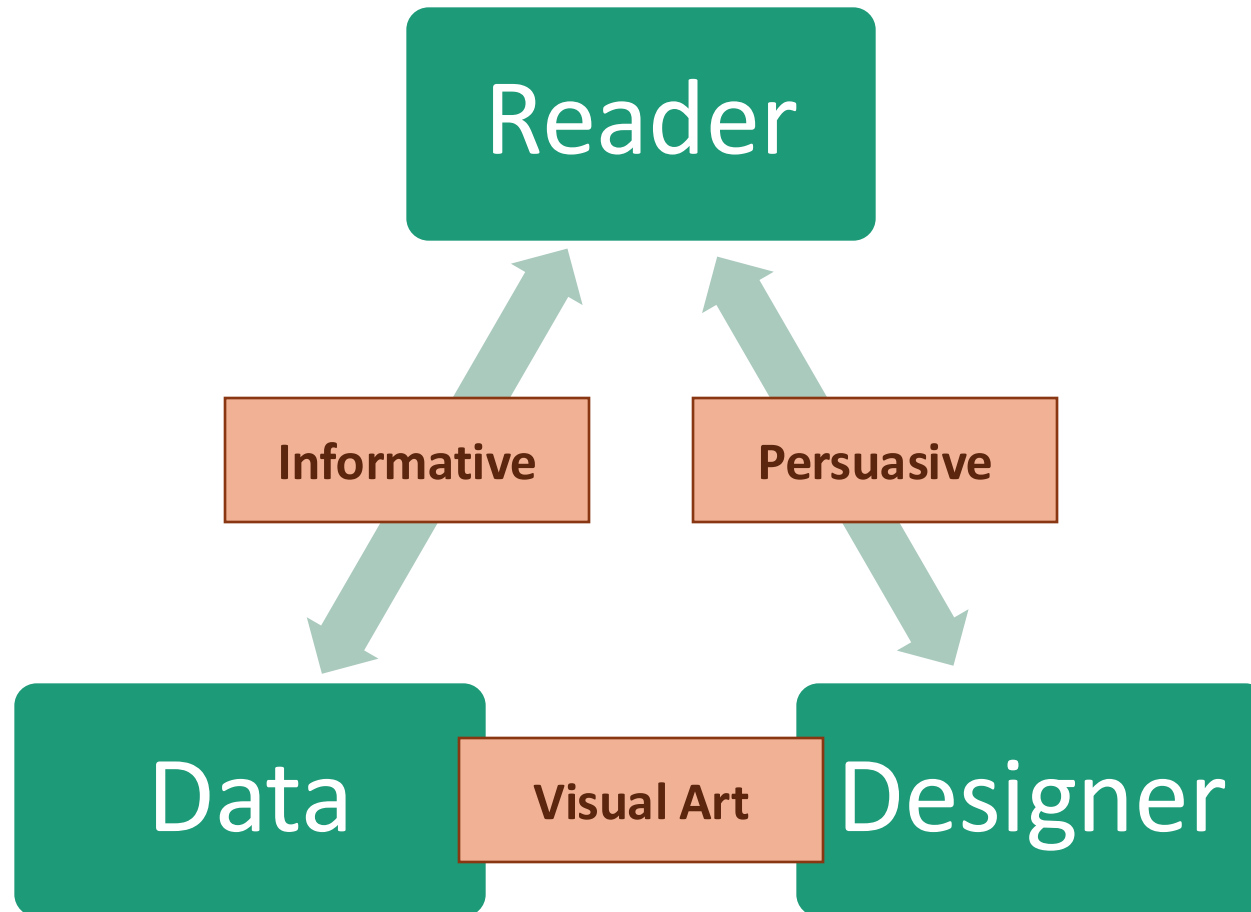
Visualisation?

*The challenge is to create **effective** and **engaging** visualizations that are appropriate to the data, **the audience and the message***

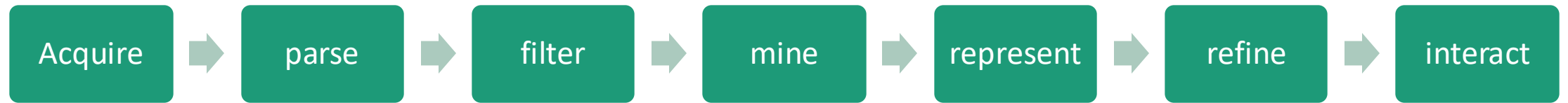
The Visualisation Trinity



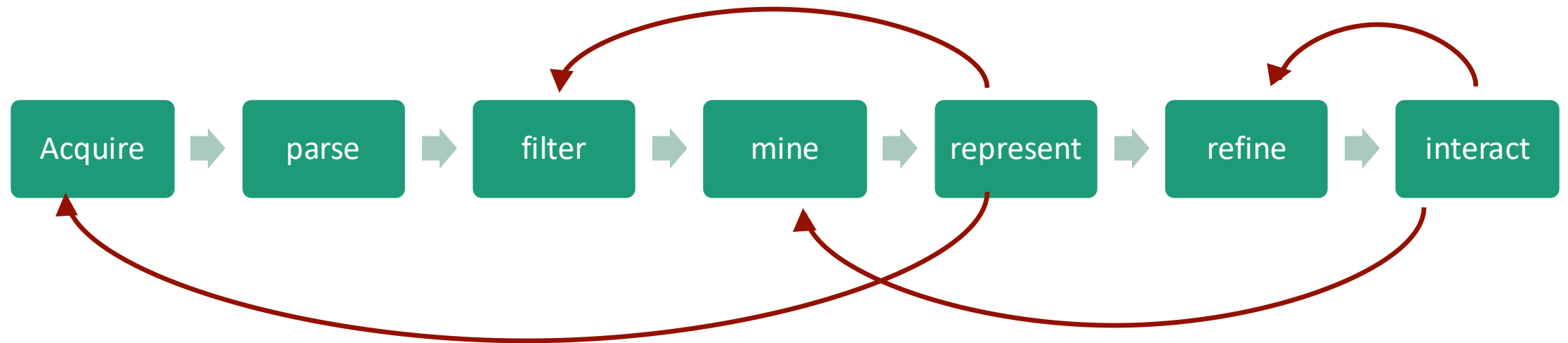
The Visualisation Trinity



Ben Fry's Visualisation Steps



Ben Fry's Visualisation Steps



Module Overview

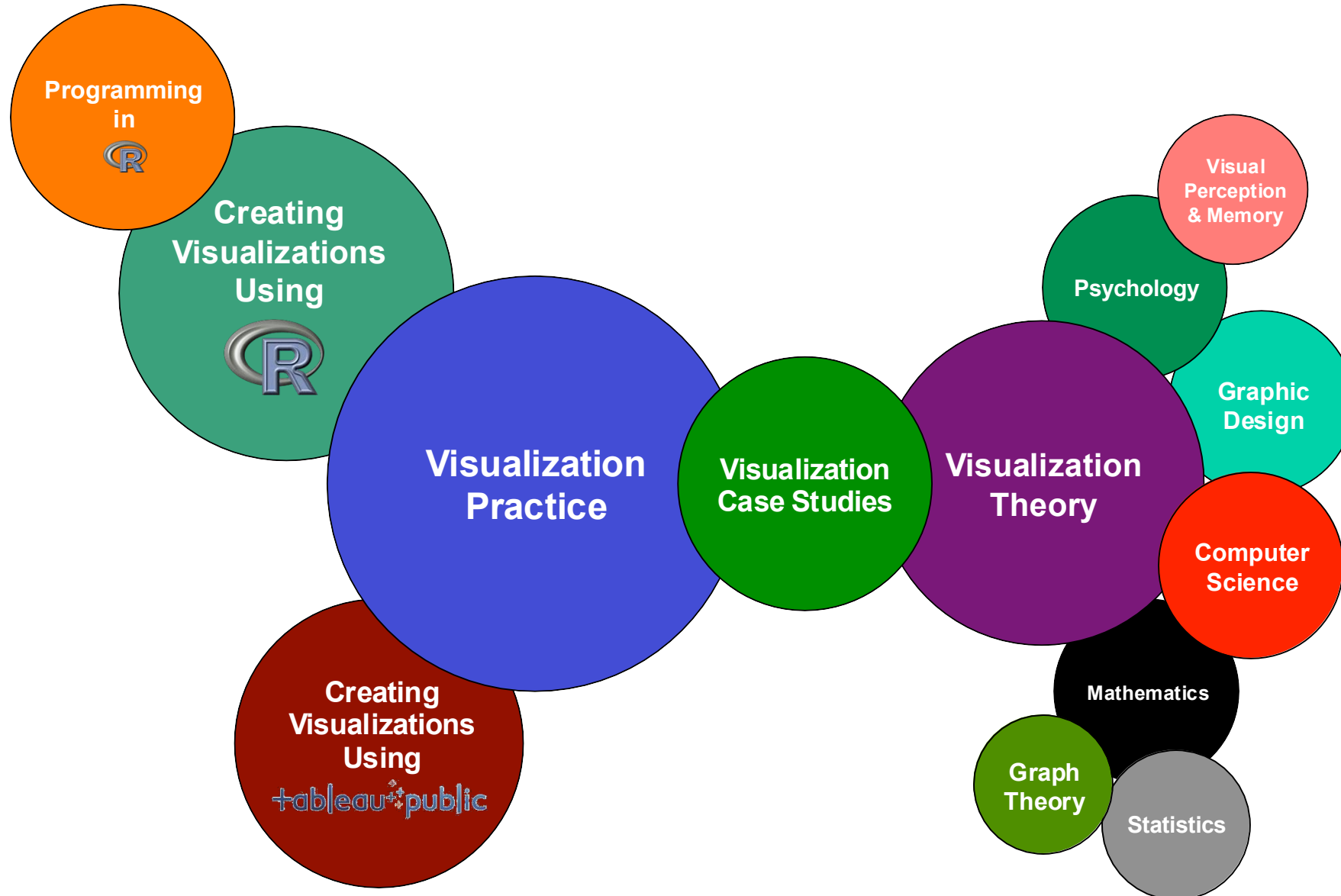
Module Aim

- The central aim of this module is help students attain the ability to take complex data, process it and extract value from it through visualisation

Learning Outcomes – as per Module Descriptor

- Demonstrate understanding of how humans perceive the world around them on a general level and absorb complex data/information on a specific level.
- Analyse and evaluate how metaphors are used to convey unfamiliar information.
- Analyse and evaluate how mental models aid in the interpretation of complex visual displays.
- Select, formulate and integrate metaphors to suit data-driven tasks
- Design effective visualizations based on principles from perceptual psychology, cognitive science, graphic design and visual art.
- Create and deploy successful data visualisations using leading software tools
- Demonstrate an understanding how visualisation is used in data journalism to communicate complex ideas and stories
- Demonstrate understanding how visualisation is used in story telling

Content



Module Content

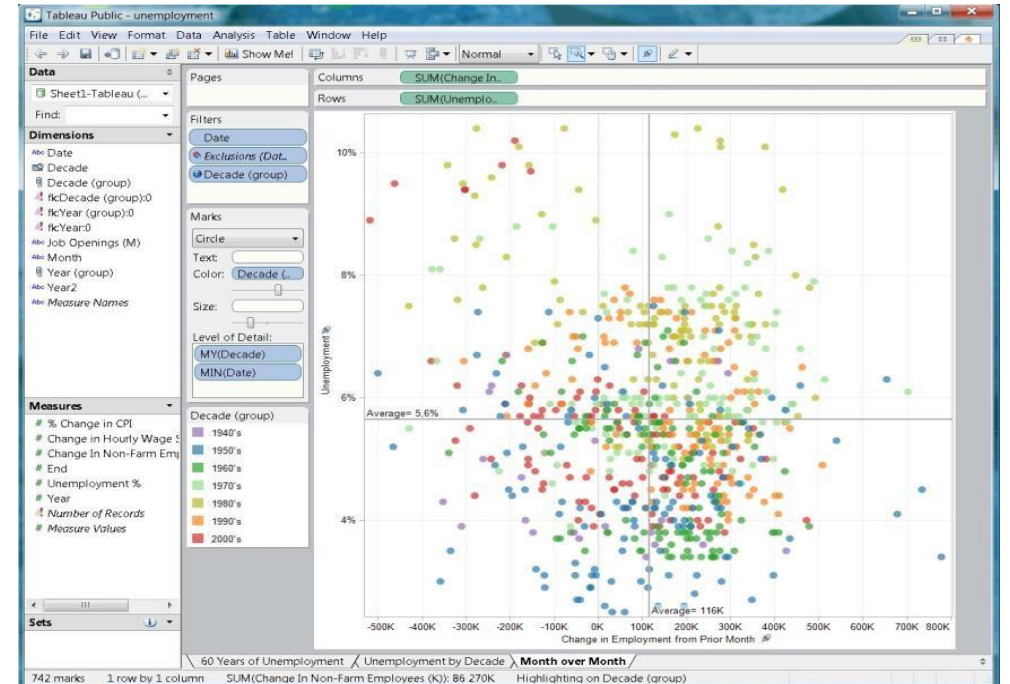
- Fundamentals of visualisation
 - Data characteristics & dimensions
 - Use of encodings
 - Colour theory
 - Graphical perception & communication
 - Interaction design
- Visualisation types:
 - Trends
 - Comparisons
 - Correlations
 - Maps
 - Networks
 - Hierarchies
 - Visualising Text
 - High- dimensional data

Tools

Tools



- Tableau is an enterprise level visualisation-based business analytics tool
- Tableau Public is its little brother!
 - <http://www.tableau.com/products/public>
- Student version:
 - <https://www.tableau.com/academic/teaching>



Tools



- R is a statistical programming
- language with extensive visualisation features
- R Studio is a rich GUI interface to R
- www.r-project.org
- www.rstudio.org

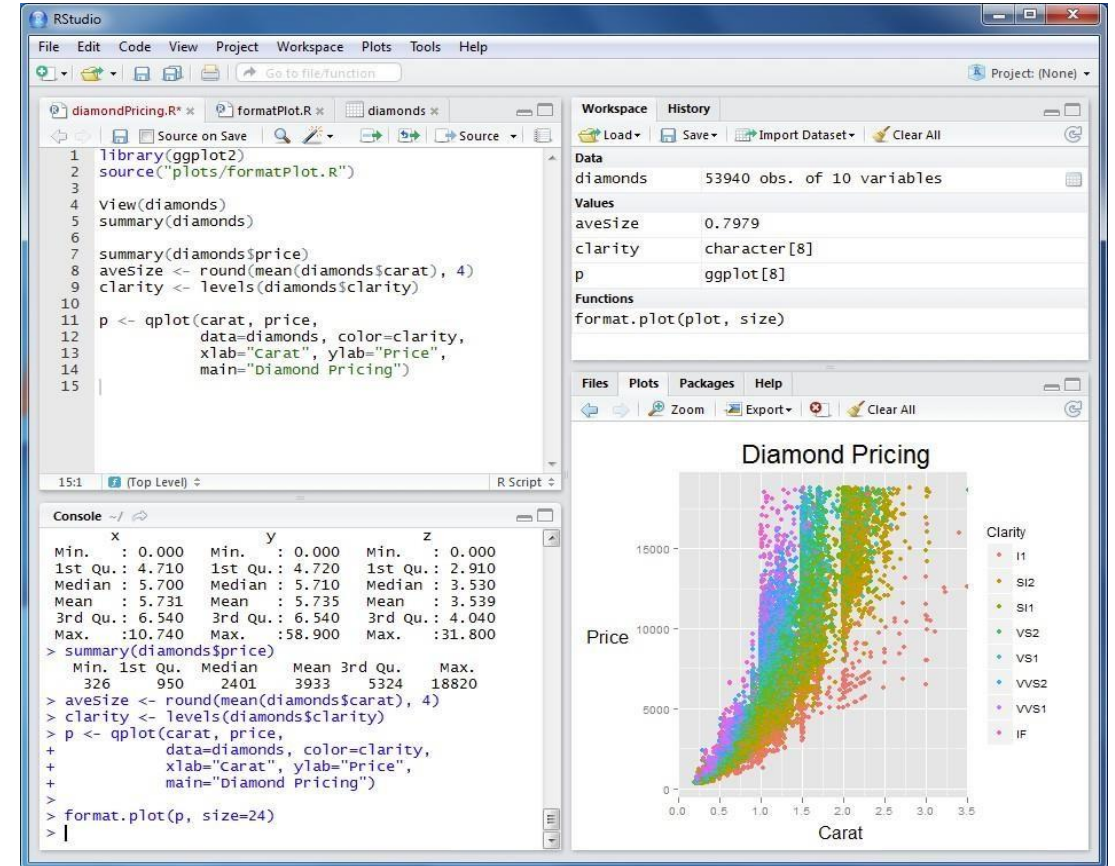


Tableau Public Visualisations

tableau⁺public

GALLERYAUTHORSBLOGRESOURCESACTIVITYSIGN IN

Gallery / Viz of the Day

Stunning data visualization examples from across the web created with Tableau Public.

Viz of the DayFeatured

step #1 UNDERSTAND

For the first time, a database maintained by the Drug Enforcement Administration tracks the path of every single pain pill sold in the United States - its manufacturer and distributor is pinpointed in every town and city - has been made public. In this view, the circles represent manufacturers that have distributed prescriptions to pharmacies in the selected city. The map is sized by quantity and colored by the type of medication that has been distributed.

step #2 SELECT A CITY

TUCSON

HOVER OVER A YEAR?

NETT SHIPPED

MANUFACTURER

PHARMACY

Visualizing the Path of Opioids

Christian Felix visualizes the path of every pain pill that enters the state of Arizona from manufacturer to pharmacy.

Featured On: August 16, 2019

A gallery of visualisations created using Tableau Public

<https://public.tableau.com/en-us/gallery>

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R Visualisations

← the R Graph
Gallery

Q CHART
TYPES

QUICK

TOOLS

ALL

D3.JS

PYTHON

DATA TO
VIZ

ABOUT



A gallery of
visualisations
created using R
with source
code!

<https://www.r-graph-gallery.com/all-graphs>

Books

- Designing Data Visualisations – Noah Iliinsky, Julie Steele (2011)
- Visualise this – Nathan Yau (2011)
- Visualising Data – Ben Fry (2008)

Also worth looking up:

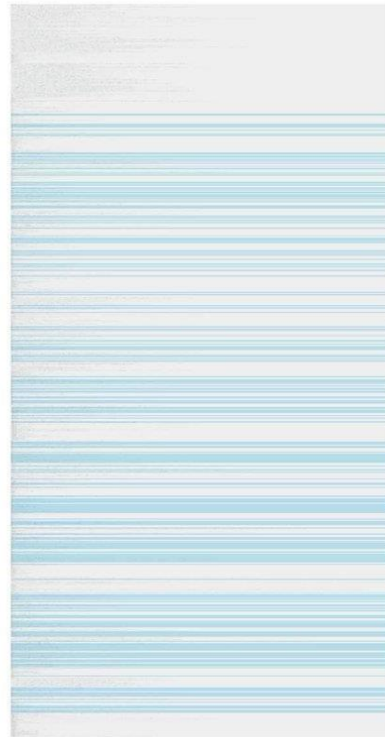
- The Visual Display of Quantitative Information – Ed Tufte (1983)
- Semiology of graphics – Jacques Bertin (1967)
- Andy Kirk – Data Visualisation (2016)

Visualization of the Week

Every time Ford and Kavanaugh did or did not answer a question

■ Answered question ■ Did not answer

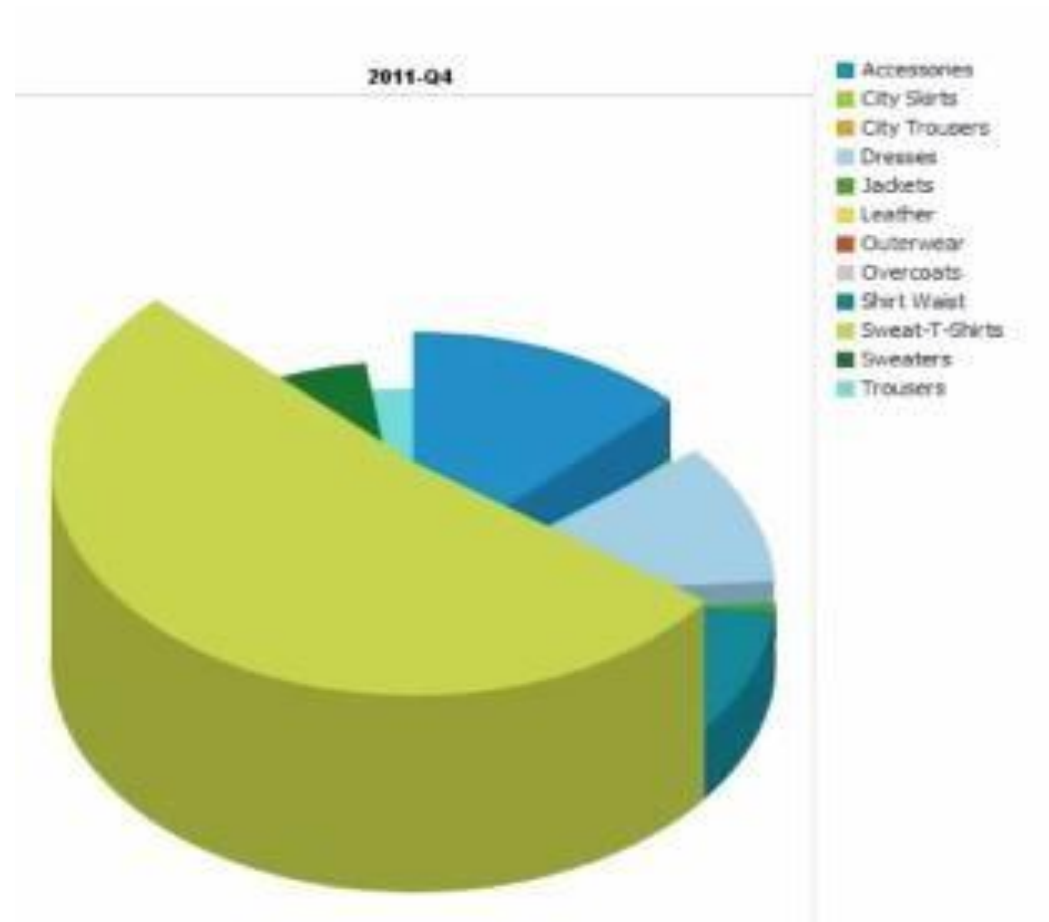
FORD



KAVANAUGH



(Un)Visualization of the Week



Remember!

- Well-designed visual representations can replace cognitive calculations with simple perceptual inferences and improve comprehension, memory, and decision making
- By making data more accessible and appealing, visual representations may also help engage more diverse audiences in exploration and analysis
- The challenge is to create effective and engaging visualizations that are appropriate to the data, the audience and the message

The Joy of Stats

- Hans Rosling
 - <https://youtu.be/jbkSRLYSojo>

For Next Time

- Find one example of good and one example of bad visualization
- Maybe visit the R and Tableau Public galleries for ideas

Thanks To

- Cathy Ennis, Marisa Llorens-Salvador, John McAuley, Colman McMahon and Brian Mac Namee for an earlier version of these lecture notes